Maju Energy Holdings

Structured Product Solution

Principal protection and upside participation for an upstream oil producer in a high volatility oil market environment

January 2026
Informational purposes only

Assumptions and disclaimers:

- The presentation to the client is conducted after the US market close on Jan 14, using market data available up to that point.
- It is assumed that the client will be able to commit the required funds and complete all legal and operational issuance procedures by 15 Jan, 9:30 EST.
- It is also assumed that the full size of all instruments involved can be traded at that time, 15 Jan 2026, 9:30 EST.
- The client is assumed to be familiar with the basics of options, futures contracts, and bond mechanics.
- Any Natixis entity hypothetically involved in the issuance process is assumed to be able to issue a zero-coupon bond at risk-free interest rates for the purpose of this product.
- Risk-free interest rates for the zero-coupon bond are assumed to be those specified in the funding grid.
- These are indicative terms only, subject to market conditions and final documentation.
- Scenarios are hypothetical and not a guarantee of performance.
- Not investment advice; suitability depends on investor circumstances.
- Structured products involve market, liquidity, and issuer credit risk.
- This presentation was prepared by a university student for a case competition and does not reflect Natixis' views or advice.

Executive Summary

Maju Energy Holdings is a Singapore-based **upstream oil exploration and production (E&P) company** holding approximately **USD 100M in treasury cash**. This cash balance supports **day-to-day operations** while providing **funding flexibility for capital-intensive exploration and development activities.**

Macro-economic Context :

1 Futures Backwardation and High Volatility Premiums

- Strong spot demand has steepened backwardation across the curve
- High implied volatility leads to call skew in the options market on oil futures

2 Geopolitical Risk is Driving Near-Term Oil Volatility

- Heightened tensions across key producing regions are driving elevated implied volatility and risk premia
- Market participants are increasingly pricing supply disruption risk into short-dated contracts

3 Supply Growth as Potential Limit for Price Gains

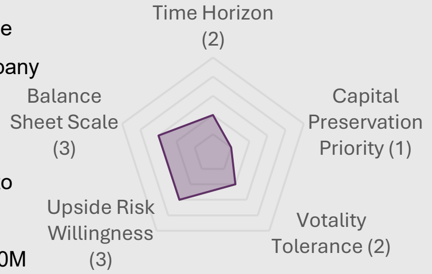
- Non-OPEC supply resilience and faster-than-expected capacity returns are expanding available barrels
- OPEC+ market share dynamics reduce upside potential despite near-term shocks

4 Energy Transition in Focus

- Slower demand growth, ESG regulation, and renewable competitiveness weigh on long-term oil consumption affecting cost, revenue and cash flow
- Near-term cash flows remain supported, but downside protection is increasingly valuable

Client Due Diligence:

1. Main Operation in Europe
2. Upstream Oil E&P Company
3. High CAPEX
4. High Carbon Emission
5. Revenue Highly Linked to Benchmark Oil Price
6. Fund Available: USD 100M



Principal Protection & Upside Potential + ESG Exposure

Our Solution: 1-year principal-protected putable structured call spread note on Brent crude oil futures to offer upside exposure while limiting downside risk and volatility

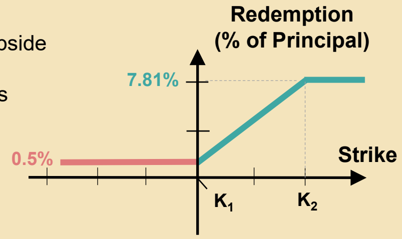
Risk

1. **Downside Risk is Structurally Limited**
   - 100% principal protection at maturity via a Zero-Coupon Bond component + 0.50% fixed coupon
   - Oil price downside does not translate into capital loss, unlike spot or futures exposure
2. **Portfolio Volatility is Significantly Reduced**
   - Product drawdown historically limited to -1.5%, versus -30%+ for spot oil
   - Downside correlation to oil prices is close to zero

Return

1. **Participation in Oil Upside with Defined Limits**
   - The call spread allows the client to capture upside
   - Returns are capped at ~7.81%

   → Provides exposure to favorable oil price moves without taking full linear risk
2. **Consistent Risk-Adjusted Return Profile**
   - Historical backtests show an average net return of ~ 2%
   - The payoff profile is positively skewed



Key ESG Impact

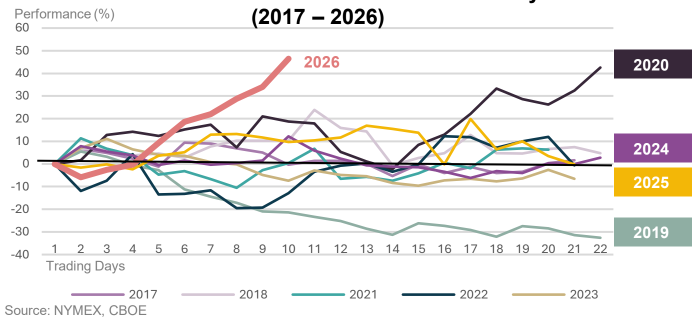
1. **Exposure to ESG Investments Will Benefit You in Four Ways:**
   - Preservation of economic optimality
   - Regulatory & stakeholder alignment
   - Provision of future financing option
   - Auditable ESG action
2. **Our Process Ensures Transparency, Accountability, and External Verification**
   - Maju invests via Natixis Structured Issuance S.A.
   - Investor funds are managed under a ESG financing framework

Market Outlook: Recent energy market behavior signals high sensitivity to geopolitical developments and uncertainty over production trends

A Historically Volatile Start of the Year…

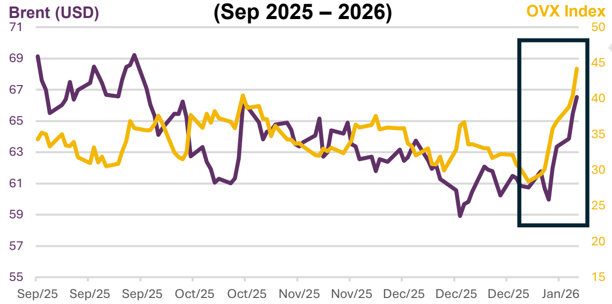
1. Steep rise of almost 50% in the **CBOE Crude Oil Volatility Index**
2. **Most volatile 2-weeks** since Isreal-Iran airstrikes in June 2025
3. Seasonals: winter months of December-January are **usually less volatile**

Historic OVD Index Performance in January (2017 – 2026)



Source: NYMEX, CBOE

Price of Brent active contract vs OVX Index (Sep 2025 – 2026)



Source: ICE Europe, CBOE

Price action has followed volatility directionally to the upside on the back of geopolitical uncertainty

1M correlation of 0.63 compared to historical slightly negative correlation of -0.1 to -0.25

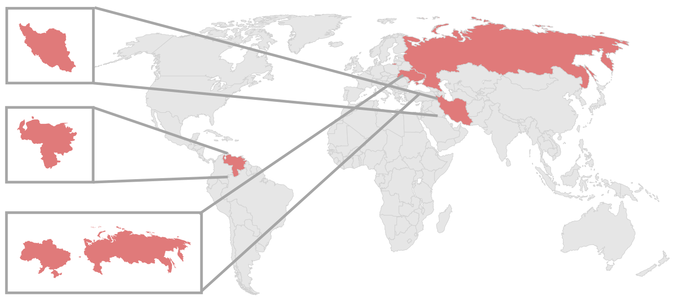
*more details available in Appendix 2.3

…Driven by Geopolitical Uncertainty…

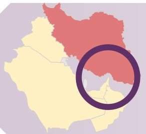
Iran's nation-wide protest and risk of US military intervention spurred fear over supply disruptions

US military actions in Venezuela heightened geopolitical risk perceptions around crude oil supply and trade routes

Russia-Ukraine war risk to the supply side is priced in, but potential for peace and lower "war premiums" act as a bearish force on price



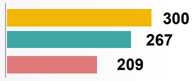
Iran's significance to global oil supply



- **Roughly 20% of global oil supply passes through the Strait of Hormuz every day**
- **A potential block of the chokepoint by Iran would disrupt this supply**

Countries with highest oil reserves (bn b):

1. Venezuela – 300
2. Saudi Arabia – 267
3. Iran – 209



Significant headline risk and unprecedented, multi-layered risk angles have markets pricing in higher volatility premiums, leading to higher oil futures and options prices and steeper backwardation

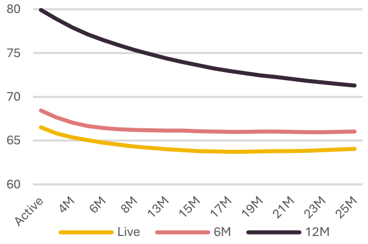
…Leading to Significant Backwardation In Brent Futures…

High demand for spot oil & short-term contracts dominate futures demand

Higher volatility premiums (& IV) due to various angles of uncertainty

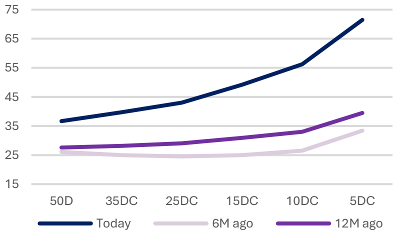
Lead to comparatively **higher options premiums** for shorter expiries

Brent Futures Curve (6M Ago – Now)



Source: ICE Europe

1M Call Skew of Brent Futures Options

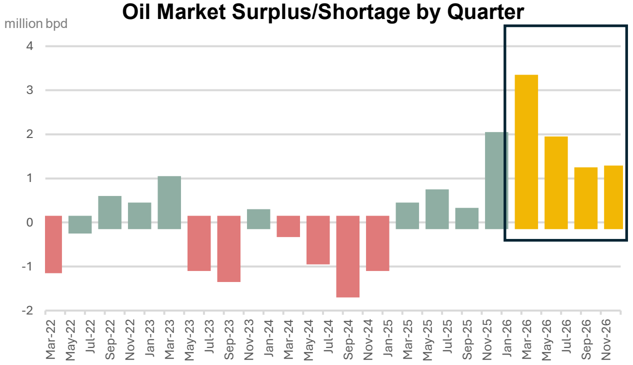


Source: ICE Futures Europe market data (via Bloomberg OVDV)

…And Uncertainty over Supply/Demand Imbalances and Production Trends

1. **OPEC+'s push for market share** has barrels returning to the market faster than expected
2. **Non-OPEC Resilience:** Countries like Brazil, Guyana, and Canada have ramped up production, adding nearly **1.3 to 1.8 million bpd** to the market
3. **U.S. Shale at "Breakeven" Edge:** Price of WTI has hovered around "pain threshold" of ~USD58 – 61 for US shale producers

Oil Market Surplus/Shortage by Quarter



Source: ING Research, IEA, EIA, OPEC

Market Outlook: Our view points to resilient near-term fundamentals alongside rising medium-term transition pressures and downside risks

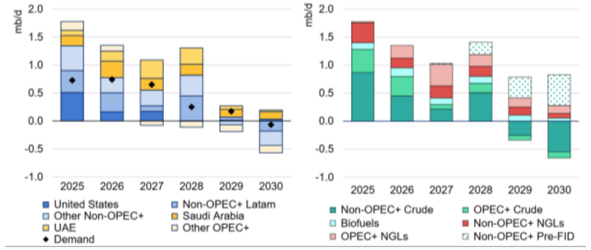
Shifting Drivers of Oil Demand and Supply Necessitate Downside Protection

…Supply Continues to Expand

Inventory buildup from excess capacity expansion: Non-OPEC+ supply growth lifts global capacity, with downside price risks emerging if production persists and inventories build

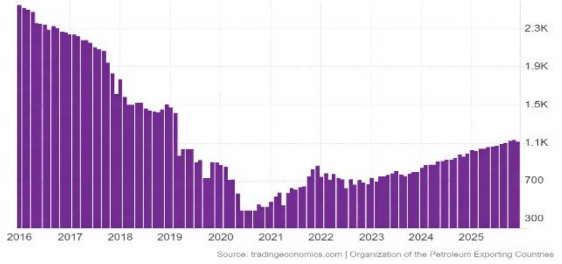
The evolving situation in Venezuela adds to U.S. control over large reserves of oil: Venezuela's production recovery could raise output to 1.3-1.4 mbd, reinforcing U.S. influence over global oil supply

Global Oil Supply Forecasts (YoY), 2025-2030



Source: IEA

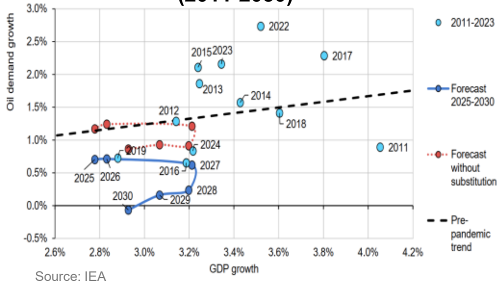
Venezuela's Oil Production (BBL/D/1K) 2016-2025



While Demand Growth is Soft …

Lower global economic growth: Oil demand tracks GDP growth and may moderate as protectionism weighs on global growth

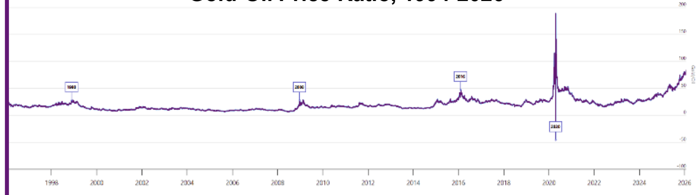
Growth in Global Oil Demand and GDP (2011-2030)



Source: IEA

The oil-gold divergence under geopolitical stress: The usual gold-oil correlation weakens in the current risk-off environment. Gold prices reflect rising political risk, while oil rallies fade amid expectations of ample supply through 2026

Gold-Oil Price Ratio, 1994-2026



Source: Yahoo Finance for COMEX Gold Price, Federal Reserve Bank of St. Louis for oil price

The green transition: Net-zero ambitions will curb oil demand, but only over the long term

While geopolitical and policy risks sustain upside volatility, **ESG-driven underinvestment supports oil price floors.**
Near-term cash flows remain resilient as energy transition impacts are **long-term, sustaining upstream revenues**
→ Strategic priorities shift toward:
Short-cycle projects to reduce transition risk **+ Downside-protected price exposure** to manage volatility and retain upside

What to Expect in the Renewables Space – Transition Risks

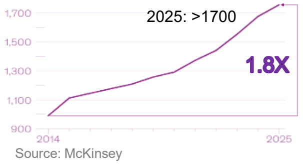
Important ESG Commitments

EU Emissions Trading System	2005
International Renewable Energy Agency	2009
The Paris Agreement	2015
U.S. Inflation Reduction Act (IRA)	2022
EU "Fit for 55"	2023
EU ESG Ratings Regulation (Upcoming)	2026

EU Carbon Permits (EUR)

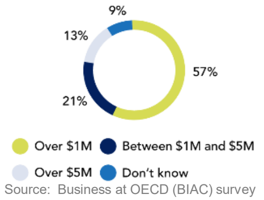
Source: Trading Economics

Number of Mandatory ESG Regulations Globally



Source: McKinsey

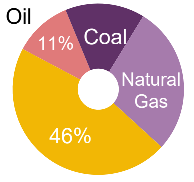
Costs of ESG reporting



Source: Business at OECD (BIAC) survey

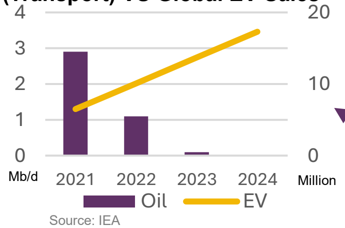
Increasingly stringent ESG regulations globally will exert upward pressure on costs, with EU carbon markets, **the world's highest pricing tier, trending upward**
→ Directly inflating production overhead

Energy Demand Growth (2024)



Source: IEA

Oil Demand Growth (Transport) VS Global EV Sales



Source: IEA

Continuous innovation and reducing LCOE are **displacing oil in the global energy mix**
→ Cheaper, scalable renewable alternatives are **driving a structural decline in long-term oil consumption**

*LCOE: Levelized Cost of Energy

Rising Demand For Renewable Energies

2017	Solar/Wind Grid Parity
2021	Global EV Inflection Point
	Over **17** Countries set target to **phase out 100% ICE** (internal combustion engines) Cars
2025	Green Hydrogen Scaling

Under tightening ESG constraints and competing demand from renewables, it is recommended to balance mandatory ESG exposure with protection and controlled upside participation

Client Overview: In view of operational cash needs and volatile margins, our priority is to safeguard your capital while keeping exposure to favorable market moves

Client Profile

Maju Energy Holdings - Singapore Oil E&P Company

Market capitalization	USD 10B
Investment Objectives	Principal Protection & Upside Exposure
Investment Horizon	6 Months to One Year
Cash Reserve	USD 100M

Business Model

Revenue and Cost Drivers:

Production Volume:
- Crude Oil
- Natural Gas
- NGLs

X

Pricing :
- Benchmark Price (Brent)

Unit Cost

… which is uncontrollable

Key Oil Fields in North Sea:

→ Your profit mainly relies on the benchmark oil price

Source: Trading Economics

Income Statement	
Company Name	**Maju Energy Holdings**
Revenue ('000)	$163,832,798.21
EBITDA ('000)	$34,327,250.49
EBIT ('000)	$16,932,185.92
Normalized EBITDA Margin**	49.00%
Normalized EBIT Margin**	32.00%
EBITDA Margin	20.95%
EBIT Margin	10.34%

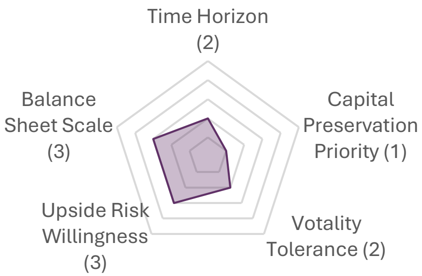
… the L1Y oil return is ~15, recent negative returns highlight the scale of downside risk

As a pure upstream producer, Maju Energy's **profitability is directly driven by benchmark oil prices**. Volatility translates into **unstable cash flows and margin risk**

***Comparable companies' reported margins are depressed by integration effects and cycle timing. For Maju Energy, a pure-play upstream client, we adopt normalized upstream margins*

Risk Appetite and Tolerance*

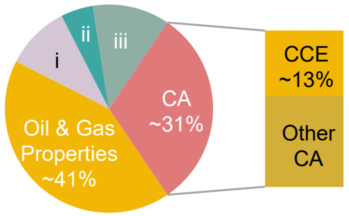
- **Time Horizon: Short-term, liquidity-constrained**
- **Capital Preservation Priority: High**
- **Volatility Tolerance (MTM): Low**
- **Upside Risk Willingness: Moderate**
- **Balance Sheet Scale: Capable**



**rated on a scale of 1 (very low risk tolerance) to 5 (very high risk tolerance)*

Total Risk Tolerance Score: 11/25 (Low-to-moderate Risk Appetite)

Asset Structure:



Debt / Equity	29.41%
Debt / Total Assets	15.14%
Net Debt / Equity	3.74%
Cash / Total Assets	13.22%
Current Ratio	1.30

Project Development Cycle:

3-8Y | 10-20Y

Exploration + Development → Production

Cash Flow from Operations ('000)	$7,047,907.01
Capex ('000)	-$4,780,110.05
Free Cash Flow ('000)	$2,267,796.96

Liquidity is Operationally Critical:

- Liquidity ratios indicate only **moderate short-term financial flexibility**
- Asset base dominated by **long-life, illiquid upstream projects** (oil fields, offshore platforms)
- Operating and development spending cannot be easily deferred

→ The cash reserve is not risk capital

Capital Intensity Drives Conservative Behavior:

- E&D requires **large capital investment**
- Ongoing **capex requirements remain high** regardless of oil prices

→ Operational priorities require **lower risk tolerance** on cash reserves

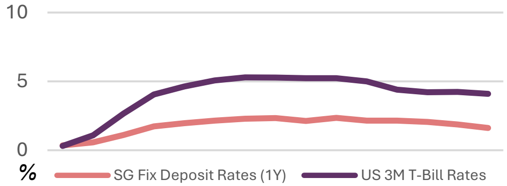
Lack of ESG Exposure:

- Upstream oil operations are **carbon-intensive**
- Tightening ESG regulations will likely increase **compliance and disclosure costs**

→ More ESG exposure is recommended to support the strategic transition

i: Exploration & Evaluation Assets, ii: PPE, iii: other non-current assets, CA: Current Assets, CCE: Cash & Cash Equivalents

Strategic Consideration



Source: MAS, USDT

While low-risk cash instruments offer **stable but small returns**, oil prices show **dynamic upside potential**

Client's Needs

1. **Protect – Principal & Liquidity**

   The client needs a product that aligns with **their low-to-moderate risk tolerance**, keeping **principal protected** and **liquidity** in focus

2. **Participate – Oil Upside & Volatility**

   The client needs a product **that protects operational cash** while **transforming oil price volatility into a strategic opportunity**

3. **Plus – ESG Exposure**

Our Product Solution: Designed to preserve capital while capturing controlled upside and optimising liquidity and capital allocation decisions

Product Overview

Our proposed solution is a **1-year principal-protected structured note** linked to **Brent crude oil** with **monthly putability from month 7 onwards**, designed to provide **controlled upside participation** via an imbedded call spread while maintaining **defined principal protection** at maturity.

1.12M **Zero-Coupon Bond**
2. Long **Call Options** on Brent futures
3. Short **Call Options** on Brent futures
*more details on product specifics on next slide and Appendix I

Product Features – Designed to Meet Your Investment Objectives

- **100% principal-protection** and min. coupon to cap your downside at a positive 0.5% return* (subject to credit risk of issuer)
- Delivering you controlled, capital-efficient **exposure to upside in oil prices** with a pre-defined cap
- Tenor of 12m and putability aligned with **operational cash needs** and secondary market liquidity risks
- A participation rate of 38% achieves your desired **lower volatility** compared to spot oil or outright futures
- Upside exposure to **volatility** in the secondary market to realize your **hedging needs** with no payoff downside

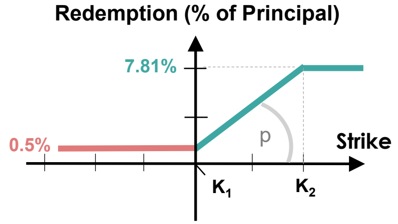
Product Details

*we express performance, strike and calculate PR in terms of Spot ($65.26 15/01, 9:30)

Issuer	Natixis Structured Issuance S.A. (S&P: A+, Fitch: A+)
Invested Principal	100M
Underlying *no principle is directly invested in futures	Brent Crude Oil Futures (ICE Brent Mar-27 – COH27)
Currency	USD
Tenor	12 months
Pr. Protection Level	100%
Min. Coupon	Fixed 0.50% (if held to maturity)
Long Call Strike	99.6% (USD 65)
Short Call Strike	118.75% (USD 77.5)
Call Options Type	European (to avoid early exercise of short call)
Part. Rate (p)	38% *calculation method on next slide
Initial Fixing	15/01/2026 at US market Open
Issue Date *10 Asian delay (Appendix)	29/01/2026
1st Putable Date	28/08/2026 (month 7 onwards)
Putability Frequ.	Monthly
Final Val. Date	28/01/2027 at US market close

- For options on futures the UL of the option is the contract that will be the front-month futures contract **at expiration time**
- **Mar-27 Brent** expires on the last business day of **January 2027**
- American style exchange options for backtesting (small difference)

Payoff Graph



Where p is the participation rate: For every 1% increase in the UL, the payoff increases by **1% * p**
(at final val. date, secondary market pricing may differ due to derivate pricing dynamics)

Payoff Formula – Final Redemption at Maturity:

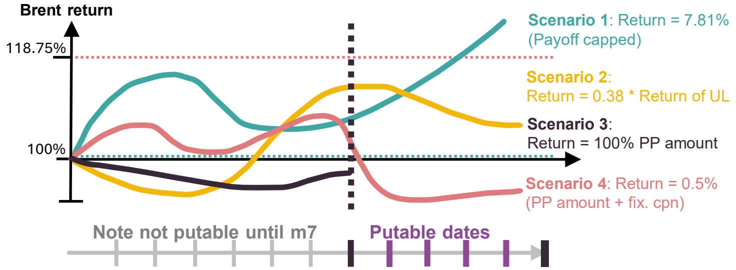
If held to maturity

$$\text{Payoff} = \begin{cases} 100\%, & S_T \leq K_1 \\ 100\% + p * ((S_T / S_0) - 1), & K_1 < S_T < K_2 \\ 100\% + p * ((K_2 / S_0) - 1), & S_T \geq K_2 \end{cases}$$

Investor put

100%

Scenario Analysis



Disclaimer:
*By Brent futures, we refer to Mar-27 Brent price of $62.71 (which will be spot Brent on the final valuation date of the product)
- Options on futures have the front-month futures contract **at expiration as an underlying, not spot**
- This price is lower than spot due to backwardation in the futures term structure, making option prices with later maturity dates appear cheaper/more favorable for the client
- We show product details as of spot as an industry norm to make the product more intuitively understandable, though we understand it improves product optics and requires a disclaimer
- Max Return = 0.5%+(p * (K_2-K_1)) * 100M = 7.808M or 7.81%

Rationale

Fundamental

- Continued **geopolitical tensions** persist as a positive tailwind for higher prices
- Rising relevance of sustainable alternatives to oil could **lower demand** for crude oil
- Higher supply, global de-escalation headline risk and softer GDP growth as **downside risk**

Technical

- Comparatively **high yields in USD** increase principal available for options premium
- **Call spread** reduces premium cost compared to an outright long call
- **Steep 1-year backwardation** on the Brent futures curve reduces call option premiums
- Debit call spread is usually slightly **net long vega** – slight volatility hedge *(MTM effect only)

Pricing the Putability Feature

Maju Energy
Corporate Treasury

We understand that tying up 100M of corporate funds comes with operational challenges and risks

And your operational exposure is long oil, demanding for liquidity options in case of price declines

1. We price a monthly Bermudan put starting at month 7 by mapping Brent futures dynamics using Black-76
2. Once the puttable window starts, we assume the put to be exercised if MTM of cont. value is below notional
3. Using backward induction of simulated paths we obtain a theoretical price of the feature

Our Pricing Considerations

- **Forward curve** of Brent futures
- The term structure of **interest rates**
- **Volatility** surface – skew and term structure, tenor/moneyness-matched
- Term structure of Natixis **default risk**
- Time to maturity, and fixed-dollar strikes

- We approximate the value to be **1% of notional**
- Assuming no FX hedge this reduces the PR from **55% to 38%**

Backtesting Methodology: We calibrate maturity, payoff structure, and currency exposure to enhance participation without compromising principal protection

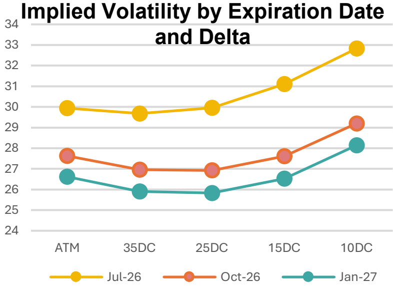
Setting Your Maturity Redemption at 100.50%

Looking at **6M**, **9M** and **12M** as possible tenors for the Zero-Coupon Bond, we see that the different total yield generated determines the funds available for derivatives exposure

we assume the yield of the ZCB to be 3.89% from the funding grid

	12M	9M	6M
Capital Needed (as % of invested principal)	96.7369	97.6510	98.5826
Capital Available (of $100M)	USD 3,263,067	USD 2,470,425	USD 1,662,661
Capital Available incl. Putability	USD 2,263,067	-	-

Call Spread Strategy – Maximizing Upside and Participation Rate



Source: ICE Futures Europe market data (via Bloomberg OVDV)

Upside skew across 12M, 9M and 6M time horizons

Long Call Rationale:
- Long ATM strike to capture upside effectively

Short Call Rationale:
- Balance high strike = high skew vs. higher credit from lower strike

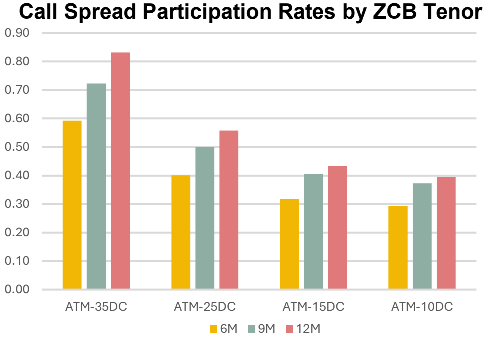
Max Payoffs Assuming 100% PR for 50D Call Spread by Short Call Delta for 12M, 9M and 6M Maturity

Option Expiry	50D	35DC	25DC	15DC	10DC
Jul-26	0	7.46%	14.42%	25.15%	34.85%
Oct-26	0	8.43%	16.12%	27.55%	38.35%
Jan-27	0	10.07%	18.68%	31.69%	44.32%

*Assumption: 50D as a proxy for ATM

- **50D-25DC** and **50D-15DC** spread upside aligns with our macro view for returns in times of geopolitical turmoil
- Next, we look at cost of the call spread structures in comparison to our available budget - this will determine the participation rate

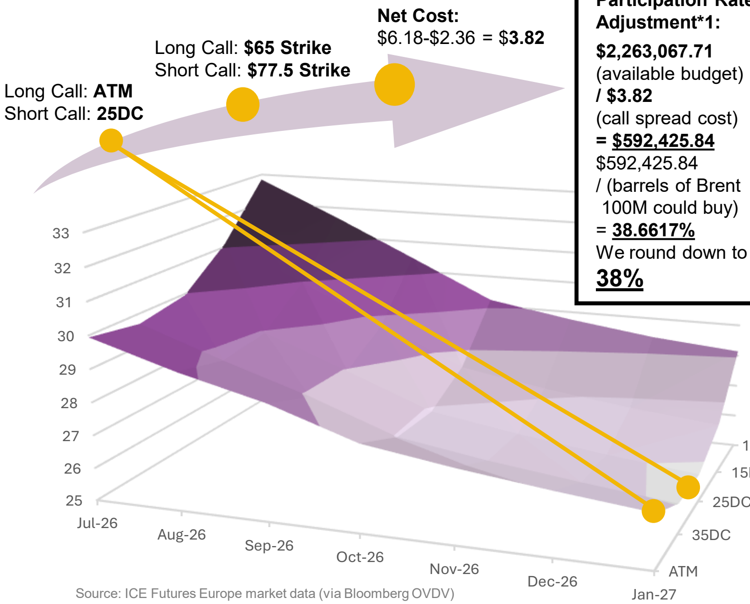
Optimizing Product Maturity for Participation



Participation rates are consistently highest for **12M** structures

Higher yield from 6M to 12M means more budget available for call spread. From 6M to 12M the price of the options increases as well due to **time value** but not to the same extent

The sweet spot of **balancing optimal tenor** while **capturing high upside** potential and **capitalizing on upside call skew**: **12M 50D-25DC Call Spread**



Participation Rate Adjustment*1:
$2,263,067.71
(available budget)
/ **$3.82**
(call spread cost)
= **$592,425.84**
$592,425.84
/ (barrels of Brent 100M could buy)
= **38.6617%**
We round down to
38%

Source: ICE Futures Europe market data (via Bloomberg OVDV)

*1we use market convention of spot price instead of underlying futures price to calculate participation rate

FX Considerations

- Our goal is to hedge oil prices and oil price volatility
- Adding an FX hedge would introduce a new source of volatility and price risk
- Your most important costs and revenue are **USD denominated**:

Drilling well construction – international USD denominated contracts
Seismic surveys and survey equipment – priced in USD
Logging, stimulation and well intervention – universally quoted in USD
~80% of crude oil sales, your main revenue source – USD-denominated

We advise to leave the structured product payoff **unhedged** in USD

...but we can offer you a hedge for balance sheet currency reporting effects

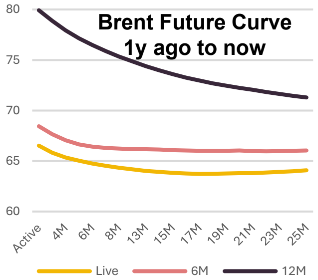
	Unhedged	9% max ccy loss	5.5% max ccy loss
PR	**38%**	**34%**	**26%**

*hedge for notional amount only, excluding potential gains from note returns

The Effect of the Futures Curve Term Structure on Option Prices

Brent Crude Oil futures trade in **backwardation** 60-70% of the time due to:
- high demand for **physical** delivery
- inelastic, controlled **supply**

In times of crisis:
- Spot prices rise faster than other contracts
- Volatility premiums increase more for spot
- The curve **steepens further into backwardation**

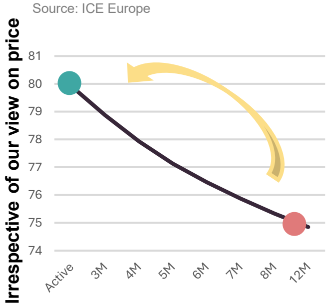


Source: ICE Europe

Effect on option prices:

Longer-dated options are struck on a **lower underlying price**	**Forward price drift** occurs gradually as expiry approaches

Our View:
- Shape of futures curve remains broadly stable through 2026 – forward price drift in your favor
- The Mar-27 contract continues to roll higher toward spot



Source: ICE Europe

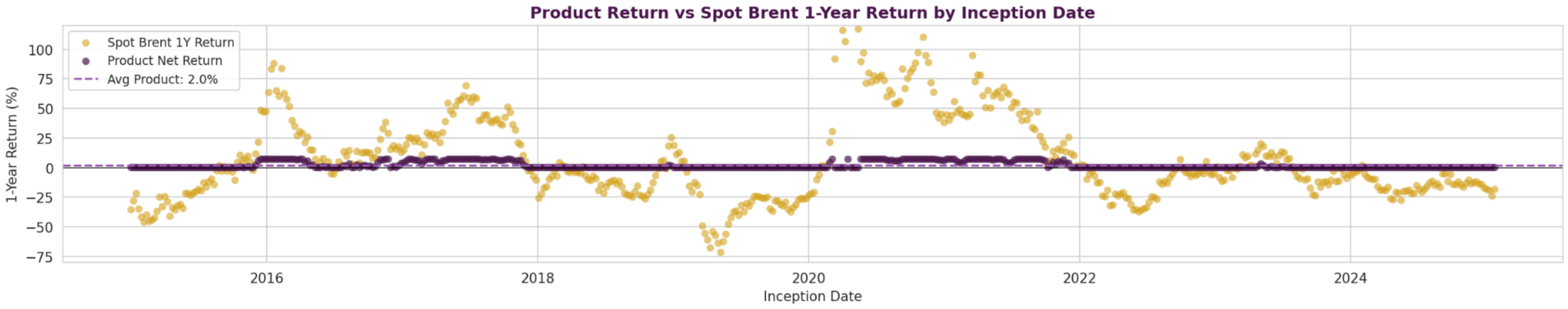
Performance Overview: Our backtests demonstrate that the structure consistently protects against drawdowns while delivering stable returns across different market cycles

Key Performance Metrics

Metric	Value
Total Backtests	526
Average Net Return	1.97%
Participation Rate	38%
Win Rate	85.7%
Total Cost	0.49%

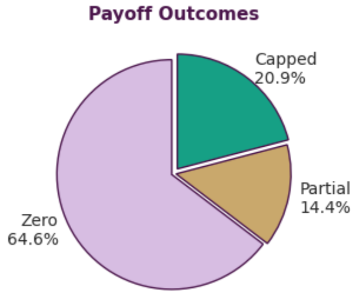
Over 10 years of rolling 1-year backtests, the structure delivered an **average 1.97% net return** while providing **100% principal protection**. The 85.7% win rate reflects periods when oil rallied above the ATM strike.

Product vs Spot Comparison



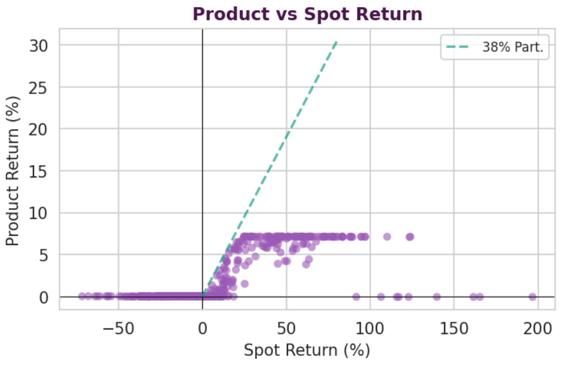
While Brent spot returns swung **between -75% and +125%,** the product's return range was constrained to -0.06% to +7.14%, demonstrating effective **downside protection**, meaningful **upside capture** and significantly **lower volatility.**

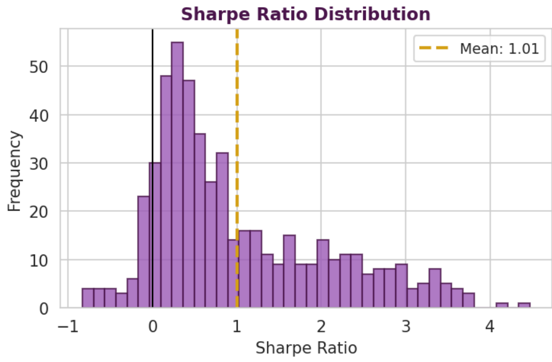
Outcome Distribution



Zero Return (64.6%)	Oil ended below strike	Principal returned in full
Partial Payoff (14.4%)	Oil rallied moderately	Partial participation captured
Capped Return (20.9%)	Oil rallied strongly	Maximum payoff achieved

Risk-Adjusted Performance: Asymmetric Payoff with Controlled Volatility





The product captures 38% of oil's upside when spot rallies (points follow the participation line), while returns floor at ~0% when spot declines. This asymmetric payoff protects principal in downturns while preserving meaningful upside exposure

With a mean **Sharpe of 1.01**, the product delivers positive risk-adjusted returns on average. The right-skewed distribution shows more frequent strong performance periods (Sharpe >1), reflecting the **principal-protected structure**

Model Explanation

For each inception date, we:

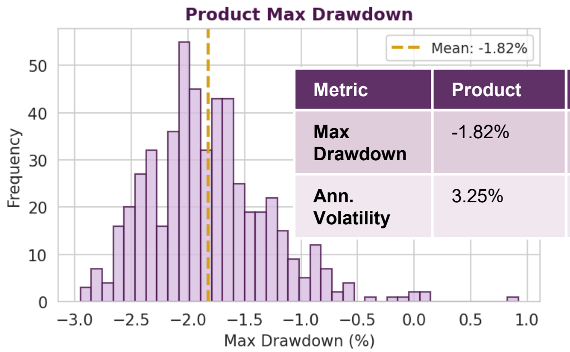
1. Calculate the ATM (50-delta) and OTM (25-delta) strikes based on that day's spot, vol, and rates
2. Price the call spread using Black-Scholes
3. Track daily mark-to-market including all Greeks
4. Apply realistic transaction costs (Natixis 1y funding at 3.89%, bid-ask, structuring fees)
5. Measure final payoff at expiry: call spread value + guaranteed 0.5% minimum return

Monte Carlo Forward-Looking Analysis

- Minimum 0.5% return (oil flat/down), principal always protected
- When oil rallies, returns can reach ~7%
- Limited downside (costs only), unlimited upside to cap
- Maju Energy needs **capital preservation first, upside participation second**

Potential Risks: Our risk assessment shows that losses are contained across adverse scenarios, with risks that are transparent and manageable

Quantitative Risk Metrics

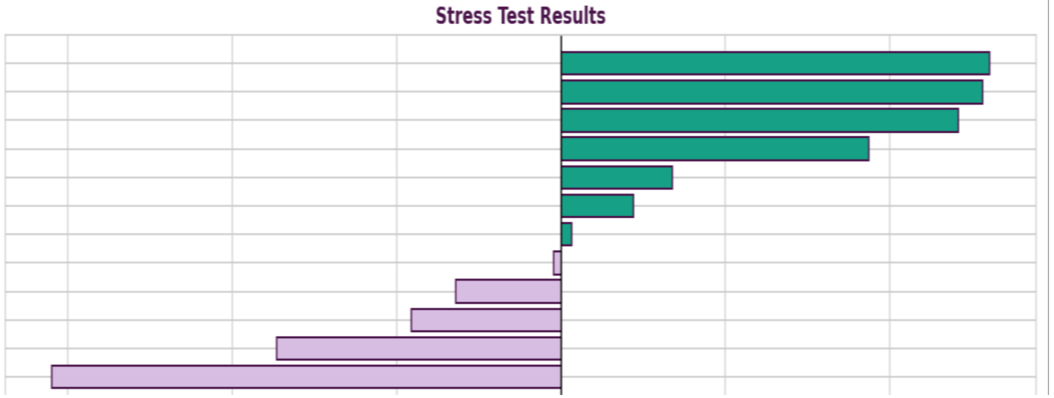


Metric	Product	Spot Brent
Max Drawdown	-1.82%	-32.81%
Ann. Volatility	3.25%	35%

Drawdown Protection: Product max drawdown of -1.82% vs Spot's -32.81% represents a 95% reduction in downside risk

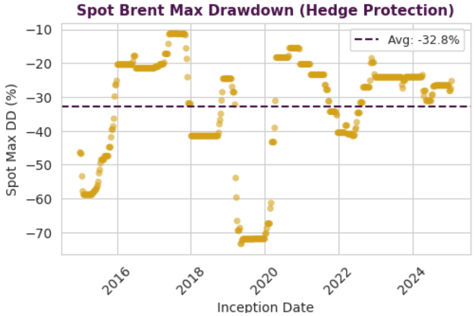
Scenario Analysis – P&L Impact

Spot Shock	P&L Impact *	Scenario
+35%	+2.61%	Strong Rally +35%
+40%	+6.57%	Ukraine War Spike
+20%	+2.42%	Moderate Rally +20%
0%	+0.68%	Vol Spike Only
0%	+0.44%	Base Case
0%	+0.07%	Vol Crush
-65%	-0.04%	COVID Crash (Mar 2020)
-20%	-0.91%	Moderate Decline -20%
-5%	-1.73%	Rate Hike +200bp
-50%	-0.64%	2014-15 Oil Crash
-40% ↑	-3.10%	Perfect Storm



Even in the worst scenario (COVID -65% crash), the product returns at least +0.5% due to the guaranteed minimum coupon

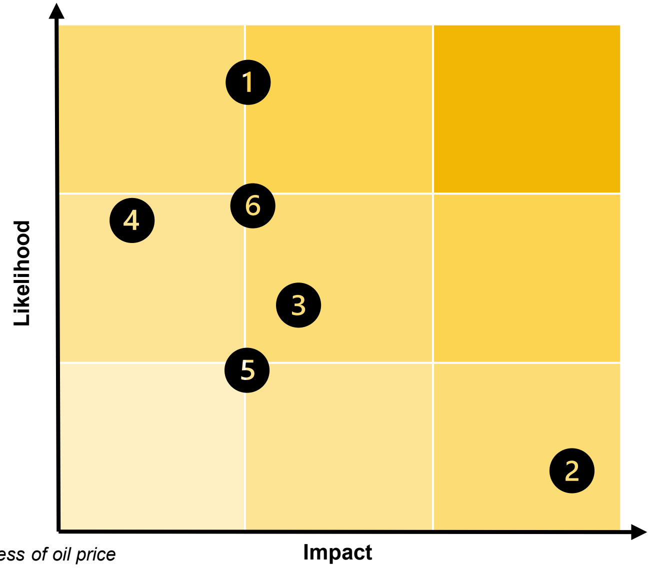
Protection Against Underlying Max Drawdown



- **Average spot oil drawdown** during holding period: -32.8%
- **Product drawdown** limited to transaction costs: -1.82%
- **Correlation to spot upside**: 0.73 (captures rallies)
- **Correlation to spot downside**: ~0 (protected)

The ZCB component ensures 100% principal return regardless of oil price movement, while the call spread captures 38% of any upside between the strikes

Key Risks & Mitigations



	Risk	Description	Mitigation
1	**Opportunity Cost**	64.6% of periods return 0% (only principal)	Acceptable given capital preservation mandate;
2	**Issuer Credit Risk**	Note depends on Natixis creditworthiness	Natixis rated A+ (S&P) / Aa3 (Moody's); consider CDS hedge if needed
3	**Liquidity Risk**	OTC structure may have limited secondary market	12-month tenor aligns with client's liquidity needs, and Natixis acts as a counterparty
4	**Vol/Rate Risk**	MTM volatility during holding period	Greeks show limited sensitivity; client holding to maturity
5	**Basis Risk**	Brent futures vs client's actual oil exposure	Brent is benchmark for client's production pricing
6	**FX Risk**	Potential balance sheet currency loss if USD/SGD depreciates	We offer imbedded option structures that cap USD downside risk while accepting a lower PR rate

** P&L shown is MTM estimate; at expiry, minimum return is guaranteed at +0.5% regardless of oil price*

ESG Considerations: Our Sustainable Note Issuance Framework Ensures that We Cater to Your Needs in an Evolving Energy Market

How ESG Investments Benefit You

Preservation of Economic Optimality: The product allows you to demonstrate ESG-aligned practices at the Treasury level while managing oil price exposure

Regulatory & Stakeholder Alignment: ESG-aligned funding supports compliance readiness amid tighter regulatory and disclosure expectations from government, credit agencies, and investors (More details on Page 3)

Provision of Future Financing Option: The structure establishes an ESG-aligned treasury precedent, creating options for future transition or sustainability-linked financing without upfront commitment

Auditable ESG Action: The arrangement provides a tangible, sustainability-aligned treasury instrument

Our Sustainable Note Issuance Framework

Economic Engine – Structures note payoff, prices note, manages options book

Maju Energy

Provides funding

Issues note

Natixis Structured Issuance S.A.

SPV and legal issuer

- Issues the note to the investor and receives cash
- Holds no market risk and acts as a pass-through entity

Passes on investor funds

Passes on call spread payoff

Natixis CIB Treasury

Issues ZCB

Passes on call spread payoff

Natixis CIB Global Markets Structuring Team

Our stucturers receive the yield of the ZCB as budget to enter the call spread strategy for your note.

Option transactions

Market

Investor funds (less yield) stay with Treasury who uses it for internal financing

Reporting & External Review

The reporting and external review framework ensures transparent use-of-proceeds tracking, measurable environmental impact, and independent verification

Annual Reporting

Allocation reporting: Details the amount of proceeds allocated, types of projects, and a summary description

Impact reporting: Quantifiable metrics such as renewable energy capacity (MW), annual energy produced (kWh), and annual GHG emissions reduced

Annual Review by External Partner Agency

Management of Proceeds

We use a **"portfolio approach"** to monitor and account for the allocation of proceeds:

- **Overall: the eligible asset base is dynamic, and the criteria are tested routinely**
- **Tracking**: A dedicated monitoring infrastructure is in place to verify and record how proceeds are allocated throughout
- **Safety Cushion**: We strive for the fact that eligible Green / Social assets exist beyond the total net Proceeds of Green & Social Financing Instruments
- **Ineligibility**: If a project becomes ineligible, proceeds are reallocated to other Eligible Green / Social Projects. Temporarily unallocated proceeds will be held in liquid forms

Eligible Uses of Funding

Eligible Green Categories

- **Green Buildings**
- **Renewable Energy**
- **Energy Efficiency**
- **Clean Transportation**
- **Information & Communications**

Eligible Social Categories

- **Affordable Housing**
- **Access to Essential Services**

Appendix Network

I. Our Quantitative Framework for Performance and Risk
   1. Pricing a Putability Feature and Visualization
   2. Pricing and Simulation for Our Product
   3. Backtesting Analysis

II. Further Product Info
   1. Indicative Term Sheet
   2. Product Lifecycle
   3. Brent-related Correlation Analysis
   4. Call Spread Analysis
   5. Issuance Wording

III. Historical Macro Market Info
   1. Historical Oil Price Analysis
   2. Volatility Analysis
   3. Rate Decisions & Market Structure

Appendix 1.1: Pricing a Putability Feature and Visualization

Pricing Model Environment: BQNT (Python)

We use a **Black-76 Monte Carlo Simulation** (100k paths) to model the **Brent futures dynamics of CO12** from value date to maturity date for each of the 6 observation dates (at each observation date it will have one month less left to maturity)

- **Calculation of Note MTM at each of the 6 reference dates:**

a. Brent futures forward
b. Brent futures volatility term structure
c. Forward curve of interest rates
d. Forward curve of issuer default risk
e. We also need the issuer's discount rate later on

$$F_{t+\Delta t} = F_t \exp\left(-\frac{1}{2}\sigma^2 \Delta t + \sigma\sqrt{\Delta t}Z\right)$$

**Martingale assumption, no drift*

No Call period of 6 months

Month 7 **Month 8** **Month 9** **Month 10** **Month 11** **Month 12**

Putable monthly at investor's discretion

…we obtain the value of the note at each putable date…

$$V_t = B_t + O_t$$

$$B_t = Ne^{-0.0389(1-t)} \qquad O_t = C(F_t, K_1, 1-t) - C(F_t, K_2, 1-t)$$

V_7 V_8 V_9 V_{10} V_{11} V_{12}

…and use backward induction to obtain the discounted price (interest rate of 3.89%) of the Bermudan Put

$$V_0^{Puttable} = \frac{1}{M}\sum e^{-rt}\, CF_i$$

Where we compare immediate exercise value (notional value amount) and the **continuation value (path dependent),** each observation date at a time using regression of the following simulated paths

Our simplified model returns

21bps = 0.21%

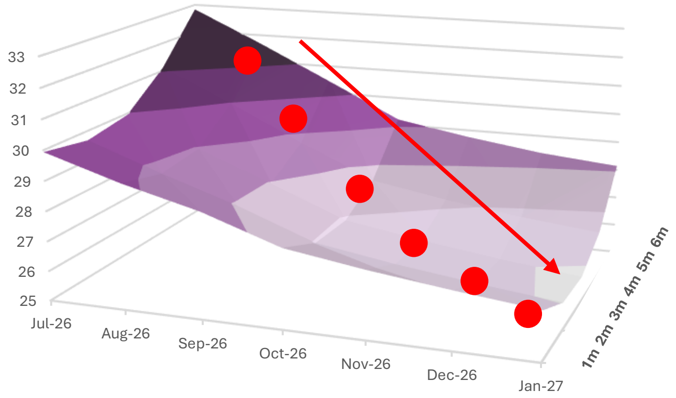
We round up to:

1%

Appendix 1.1: Pricing a Putability Feature and Visualization (cont.)

The Price We Seek: our expected cost of granting the **right to put** the note to the investor

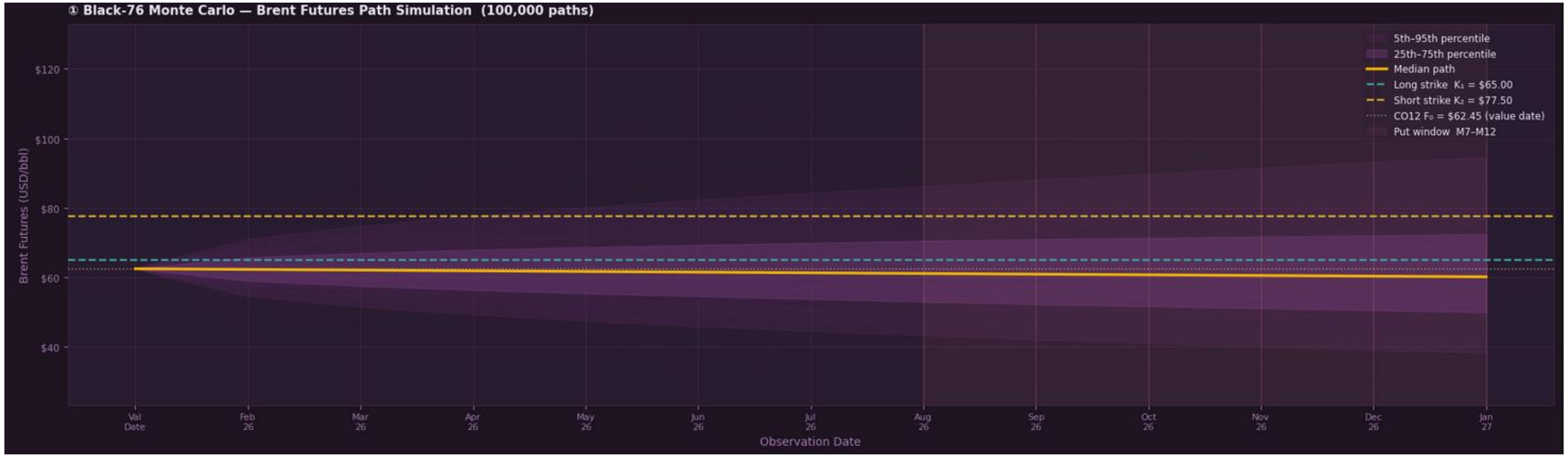
The Problem: multiple exercise dates with a path-dependent underlying

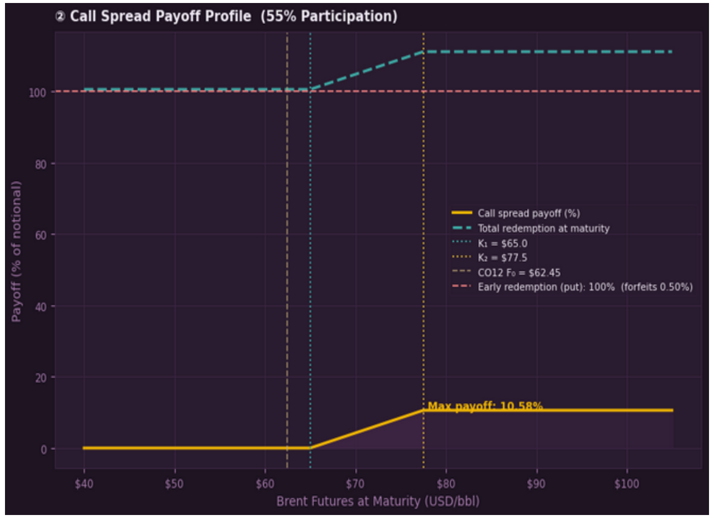
Qualitative Pricing Considerations	How we use it in the Model – data described below assumed to be collected on pricing date before note inception	Example
1 Forward curve of Brent oil futures – specifically (ICE Brent Mar-27 – COH27) • Affects the MTM mapping of underlying futures value	For each of the 5 future observation dates we take the forward price of the contract that has **the remaining time to maturity of the note as the time to expiry to match**	For putability date 28/08/26 we use the forward price of the future that on that day will have 5 months left to expiry
2 Term structure of interest rates – the issuer discount curve • The effective interest rate for the remainder of the note will affect ZCB and options value	For each future putability reference date we take the forward price of the rate that on that day has the same tenor as our note	On the date in Oct the note has a time to maturity of 3 months – so we take the forward rate of the 3m yield in Oct
3 Volatility Surface of Brent futures • Skew and the term structure of volatility will affect how we price the options for each of the putability dates which has a huge pricing effect with multi-dimensional risks	For each future putability reference date we take the implied volatility of the option which, on that date, has the time tenor of the remainder of the note	
4 Term Structure of Issuer Credit Risk • If the price of Natixis CDS changes over time/default risk becomes larger or smaller, this will affect the value of the ZCB quantitatively and the continuation value qualitatively	For each future putability reference date we take the current forward price of the CDS that will have a tenor equal to the remaining maturity of our note	

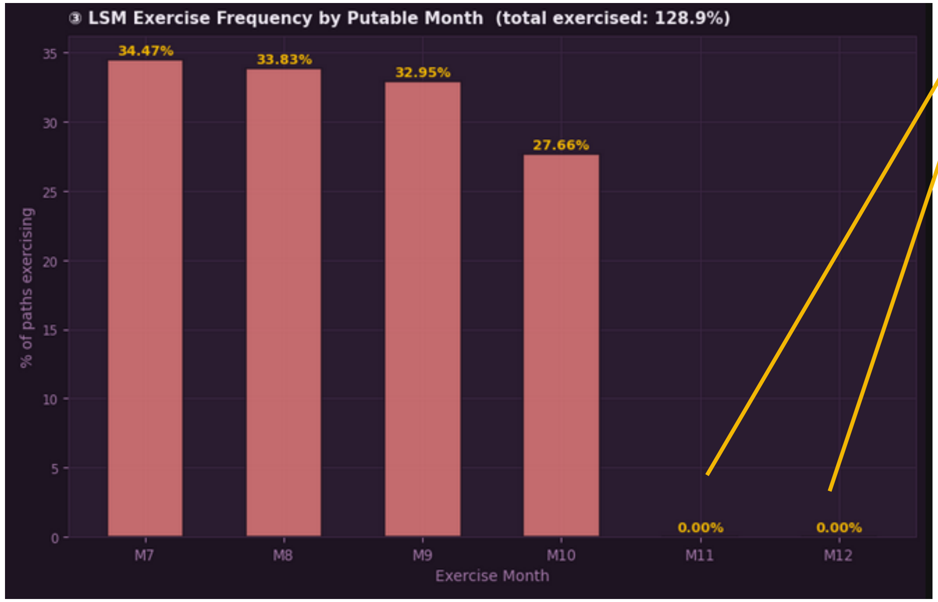
5 The Strikes remain constant and time to maturity increase intuitively with Maturity date – putability date

We have to assume to be able to get accurate prices and market implied rates and prices for each of the above-mentioned aspects

Appendix 1.1: Pricing a Putability Feature and Visualization (cont.)







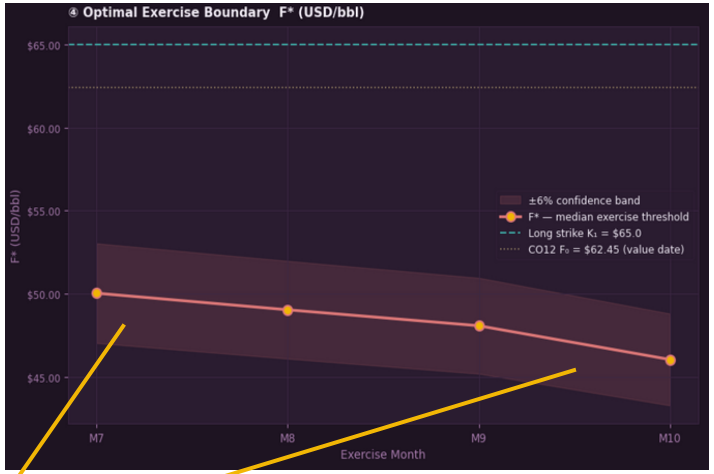
We can see the 0.50% fixed coupon at maturity becoming increasingly attractive as we approach maturity (the feature as **absolutely no value** here)

Put Value: 21 bps

Limitations: Didn't include CDS curve for Natixis, no strike specific vol profile to avoid Black-76 inconsistencies, Black-76 isn't perfect

We are more comfortable to round up to: 1%

And the investor's futures cut-off price falls as maturity date approaches for the same reason and lower continuation value



Appendix 1.1: Pricing a Putability Feature and Visualization (cont.)

Parameter	Value	Result	Detail
Product	1-Year Principal Protected Note · Brent Crude	Put Value	0.2141% ≡ 21.41 bps
Value Date	15 Jan 2026	PV (no put)	99.8478% of notional
Principal Prot.	100.50% at maturity	PV (with put)	100.0619% of notional
Call Spread	Long 65.00/*Short*77.50	Max CS Payoff	10.5769% of notional
Participation	55%	Simulation	M = 100,000 paths · Black-76 · Antithetic variates
CO12 F_0 (Trade Date)	$62.45 / bbl (CO12 Comdty)	LSM Regression	Degree-3 polynomial · ITM paths only
Put Window	Monthly · Months 7 – 12	Redemption	100.00% on early exercise (forfeits 0.50% coupon)
12M vol $\sigma(T_{12})$	27.6% p.a. (COIV12M)	OIS rate (1Y)	3.50% p.a.

```
VALUE_DATE      = date(2026, 1, 15)
MATURITY_MONTHS = 12
PRINCIPAL       = 100.0
PRINCIPAL_PROT  = 100.50

STRIKE_LOW      = 65.00
STRIKE_HIGH     = 77.50
PARTICIPATION   = 0.55
F0_SPOT_CO1     = 65.26

PUT_EXERCISE_MONTHS = [7, 8, 9, 10, 11, 12]
REDEMPTION_ON_PUT   = PRINCIPAL

N_PATHS         = 100_000
N_STEPS         = MATURITY_MONTHS
SEED            = 42

LSM_POLY_DEGREE = 3
```

```
FUTURES_CURVE = np.array([
    65.35, 64.69, 64.30, 64.04, 63.76, 63.61,
    63.46, 63.19, 63.11, 63.20, 62.99, 62.45
])
```

```
ATM_VOLS_PCT = np.array([
    47.85, 43.04, 39.04, 35.86, 34.24, 32.84,
    31.32, 30.57, 29.50, 28.78, 28.1, 27.55
])
```

```
SWAP_RATES_PCT = np.array([
    3.67105, 3.67828, 3.66913, 3.66045, 3.65005, 3.62922,
    3.60615, 3.58445, 3.5609, 3.5375, 3.51598, 3.49803
])
```

```
ATM_VOLS       = ATM_VOLS_PCT  / 100.0
SWAP_RATES     = SWAP_RATES_PCT / 100.0
T_VEC          = np.arange(1, 13) / 12.0
DISCOUNT_FACS = np.exp(-SWAP_RATES * T_VEC)
F0_SPOT        = FUTURES_CURVE[-1]
```

```
assert len(FUTURES_CURVE) == 12, "FUTURES_CURVE must have exactly 12 elements"
assert len(ATM_VOLS)      == 12, "ATM_VOLS_PCT must have exactly 12 elements"
assert len(SWAP_RATES)    == 12, "SWAP_RATES_PCT must have exactly 12 elements"
print("✓ Market data validated — 12 pillars loaded for all three inputs.")
print(f"  Futures curve  : {FUTURES_CURVE[:4].round(2)} ... (M1–M4)")
print(f"  ATM vols       : {ATM_VOLS_PCT[:4].round(1)} % ... (M1–M4)")
print(f"  Swap rates     : {SWAP_RATES_PCT[:4].round(2)} % ... (M1–M4)")
print(f"  Discount facs  : {DISCOUNT_FACS[:4].round(4)} ... (M1–M4)\n")
```

Appendix 1.1: Pricing a Putability Feature and Visualization (cont.)

```python
def simulate_black76(futures_curve, atm_vols, n_paths, n_steps, seed=42):

    np.random.seed(seed)
    dt = 1.0 / 12.0
    half = n_paths // 2

    Z      = np.random.standard_normal((half, n_steps))
    Z_full = np.concatenate([Z, -Z], axis=0)   # antithetic pairs

    F0_underlying = futures_curve[-1]    # CO12 Comdty PX_LAST  (index 11)
    sigma_12m     = atm_vols[-1]         # COIV12M Comdty vol   (index 11)

    paths        = np.zeros((n_paths, n_steps + 1))
    paths[:, 0] = F0_underlying          # initialise at CO12 price, NOT CO1

    for t in range(n_steps):
        # Constant 12M vol — same contract evolving through time
        paths[:, t + 1] = paths[:, t] * np.exp(
            -0.5 * sigma_12m**2 * dt + sigma_12m * np.sqrt(dt) * Z_full[:, t]
        )

    return paths
```

```python
def call_spread_payoff(F_T, strike_low, strike_high, participation):
    """
    Capped call spread payoff, normalised as % of notional.

    Payoff(F_T) = Participation × max( min(F_T, K₂) − K₁, 0 ) / K₁ × 100
    """
    intrinsic  = np.clip(F_T - strike_low, 0.0, strike_high - strike_low)
    return participation * intrinsic / strike_low * PRINCIPAL
```

```python
def lsm_bermudan_put(paths, discount_facs, atm_vols,
                     put_months, redemption_value,
                     strike_low, strike_high, participation,
                     poly_degree=3):

    n_paths, n_cols = paths.shape
    n_steps         = n_cols - 1

    # —— Helper: analytical note hold value at month t ————————————————————————
    def black76_call_vec(F, K, sigma, T, df):
        """Vectorised Black-76 call price (% of notional)."""
        if T <= 1e-8 or sigma <= 1e-8:
            return df * np.maximum(F - K, 0.0) / strike_low * PRINCIPAL * participation
        d1 = (np.log(F / K) + 0.5 * sigma**2 * T) / (sigma * np.sqrt(T))
        d2 = d1 - sigma * np.sqrt(T)
        return df * (F * norm.cdf(d1) - K * norm.cdf(d2)) / strike_low * PRINCIPAL * participation

    def note_hold_value(F_t, month_t):
        """
        Mark-to-market of the note at month_t for a vector of futures prices F_t.
        Uses analytical Black-76 pricing for the remaining call spread value.
        """
        remaining_T = (n_steps - month_t) / 12.0
        # Forward discount factor from month_t to maturity
        df_rem  = discount_facs[n_steps - 1] / (discount_facs[month_t - 1] if month_t > 0 else 1.0)
        vol_rem = atm_vols[-1]         # always use 12M vol — same CO12 contract throughout

        long_call  = black76_call_vec(F_t, strike_low,  vol_rem, remaining_T, df_rem)
        short_call = black76_call_vec(F_t, strike_high, vol_rem, remaining_T, df_rem)
        cs_value   = np.maximum(long_call - short_call, 0.0)

        pv_principal = PRINCIPAL_PROT * df_rem
        return pv_principal + cs_value

    # —— Initialise: terminal payoff at month 12 ————————————————————————————————
    exercise_time     = np.full(n_paths, n_steps, dtype=int)
    exercise_freq     = {}
    exercise_boundary = {}

    F_T            = paths[:, n_steps]
    terminal_payoff = PRINCIPAL_PROT + call_spread_payoff(F_T, strike_low, strike_high, participation)
    path_values    = terminal_payoff.copy()

    # —— Backward induction across exercise dates ———————————————————————————————
    for month in sorted(put_months, reverse=True):
        F_now    = paths[:, month]
        df_this  = discount_facs[month - 1]

        # Identify ITM paths: hold value < redemption (investor has incentive to consider exercise)
        hold_val = note_hold_value(F_now, month)
        itm      = hold_val < redemption_value

        if itm.sum() > 100:
            X     = F_now[itm]
            basis = np.column_stack([X**p for p in range(poly_degree + 1)])
            # Y = continuation value discounted to this month
            Y     = path_values[itm] * (discount_facs[n_steps - 1] / df_this)
            coeffs, _, _, _ = np.linalg.lstsq(basis, Y, rcond=None)
            basis_all  = np.column_stack([F_now**p for p in range(poly_degree + 1)])
            cont_value = basis_all @ coeffs
        else:
            cont_value = hold_val   # fallback: use analytical hold value

        # Exercise decision
        immediate    = np.full(n_paths, redemption_value)
        exercise_now = itm & (immediate > cont_value)

        exercise_time[exercise_now]  = month
        path_values[exercise_now]    = redemption_value

        exercise_freq[month] = exercise_now.sum() / n_paths * 100
        exercise_boundary[month] = (
            np.percentile(F_now[exercise_now], 50) if exercise_now.sum() > 0 else np.nan
        )

        print(f"  Month {month:2d}: {exercise_now.sum():6,d} paths exercise "
              f"({exercise_freq[month]:.2f}%)  |  "
              f"F* = ${exercise_boundary[month]:.2f}"
              if not np.isnan(exercise_boundary[month])
              else f"  Month {month:2d}: {exercise_now.sum():6,d} paths exercise "
                   f"({exercise_freq[month]:.2f}%)  |  F* = n/a (no exercise)")

    # —— Discount each path's cash flow to t=0 ——————————————————————————————————
    pv_cashflows = np.array([
        path_values[i] * discount_facs[exercise_time[i] - 1]
        for i in range(n_paths)
    ])

    # Baseline: note value with no put (all paths held to maturity)
    baseline_cf = PRINCIPAL_PROT + call_spread_payoff(
        paths[:, n_steps], strike_low, strike_high, participation
    )
    pv_baseline = np.mean(baseline_cf * discount_facs[n_steps - 1])
    pv_with_put = np.mean(pv_cashflows)
    put_value_pct = pv_with_put - pv_baseline

    print(f"\n  PV (no put)     : {pv_baseline:.4f}%")
    print(f"  PV (with put)   : {pv_with_put:.4f}%")
    print(f"  Put Value       : {put_value_pct:.4f}%  ({put_value_pct * 100:.2f} bps)")

    return put_value_pct, exercise_freq, pv_cashflows, exercise_boundary, pv_baseline, pv_with_put
```

Appendix 1.1: Pricing a Putability Feature and Visualization (cont.)

Main execution

```python
if __name__ == "__main__":

    print("=" * 68)
    print("  BERMUDAN PUT PRICER — Black-76 Longstaff-Schwartz Monte Carlo")
    print("  1Y 100.50% PP Note  |  Brent Call Spread  |  Monthly Put M7–M12")
    print("=" * 68)

    # ── Step 1: Simulate paths ────────────────────────────────────────────
    print(f"\n[1/3] Simulating {N_PATHS:,} Black-76 paths (antithetic variates)...")
    paths = simulate_black76(FUTURES_CURVE, ATM_VOLS, N_PATHS, N_STEPS, SEED)
    print(f"  Path matrix  :  {paths.shape[0]:,} paths  ×  {paths.shape[1]} time steps")
    print(f"  Median F(T)  :  ${np.median(paths[:, -1]):.2f}/bbl  at month {N_STEPS}")
    print(f"  95th pctile  :  ${np.percentile(paths[:, -1], 95):.2f}/bbl")
    print(f"  5th  pctile  :  ${np.percentile(paths[:, -1], 5):.2f}/bbl")

    # ── Step 2: Run LSM ───────────────────────────────────────────────────
    print("\n[2/3] Running Longstaff-Schwartz backward induction...")
    print(f"  Exercise dates      : months {PUT_EXERCISE_MONTHS}")
    print(f"  Redemption on put   : {REDEMPTION_ON_PUT:.2f}% of notional\n")

    (put_value_pct, exercise_freq, pv_cashflows,
     exercise_boundary, pv_baseline, pv_with_put) = lsm_bermudan_put(
        paths           = paths,
        discount_facs   = DISCOUNT_FACS,
        atm_vols        = ATM_VOLS,
        put_months      = PUT_EXERCISE_MONTHS,
        redemption_value= REDEMPTION_ON_PUT,
        strike_low      = STRIKE_LOW,
        strike_high     = STRIKE_HIGH,
        participation   = PARTICIPATION,
        poly_degree     = LSM_POLY_DEGREE
    )

    # ── Step 3: Print summary ─────────────────────────────────────────────
    print("\n" + "=" * 68)
    print("  PRICING RESULTS")
    print("=" * 68)
    print(f"  Bermudan Put Value  :  {put_value_pct:.4f}%  of notional")
    print(f"                      :  {put_value_pct * 100:.2f} bps")
    print(f"  Note PV (no put)    :  {pv_baseline:.4f}%")
    print(f"  Note PV (with put)  :  {pv_with_put:.4f}%")
    print(f"\n  Exercise summary:")
    for m in sorted(exercise_freq.keys()):
        f_star = exercise_boundary.get(m, float("nan"))
        f_str  = f"${f_star:.2f}/bbl" if not np.isnan(f_star) else "n/a"
        print(f"    Month {m:2d}  →  {exercise_freq[m]:6.3f}% of paths  |  F* = {f_str}")
```

Appendix 1.2: Quantitative Framework for Pricing and Simulation

Data Inputs

Data	Source	Ticker	Range
Brent Spot	Bloomberg	CO1 Comdty	2015-2026
1Y ATM Implied Vol	Bloomberg	BRCV1Y Index	2015-2026
1Y US Treasury	Bloomberg	USGG12M Index	2015-2026

Key Assumptions

1. Options priced using Black-Scholes (European exercise)
2. No early redemption or secondary market trading
3. Issuer (Natixis) does not default
4. Funding at Natixis' 1Y USD rate (3.89%)
5. Brent futures as underlying (not spot)
6. All costs deducted linearly over holding period

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```python
@dataclass
class ProductConfig:
    client_name: str = 'Maju Energy Holdings'
    notional: float = 100_000_000
    currency: str = 'USD'
    tenor_months: int = 12
    tenor_days: int = 252
    trade_date: str = '2026-01-15'
    underlying: str = 'ICE Brent Crude'
    long_call_delta: float = 0.50
    short_call_delta: float = 0.25
    target_participation: float = 0.38
    use_fixed_participation: bool = True
    protection_level: float = 1.00
```

```python
@dataclass
class CostConfig:
    issuer_funding_rate: float = 0.0389
    option_bid_ask_pct: float = 0.015
    broker_commission: float = 1.50
    clearing_fee: float = 0.50
    lot_size: int = 1000
    slippage_bps: float = 3.0
    structuring_fee_bps: float = 25
    distribution_fee_bps: float = 15
    cva_charge_bps: float = 5.0

@dataclass
class BacktestConfig:
    sample_frequency: int = 5
    mc_paths: int = 10000
    var_confidence: float = 0.95

product = ProductConfig()
costs = CostConfig()
bt_cfg = BacktestConfig()
```

Product Structure

Component	Description
Zero-Coupon Bond	Provides 100% principal protection at maturity
Bull Call Spread	Long 50-Delta call + Short 25-Delta call on Brent Crude
Participation	38% of oil upside between strikes
Tenor	12 months

Cost Structure

Cost Component	Value	Formula
Issuer Funding	3.89%	Natixis 1Y USD rate
Bid-Ask Spread	1.50%	Premium × 1.5%
Structuring Fee	25 bps	Notional × 0.25%
Distribution Fee	15 bps	Notional × 0.15%
CVA Charge	5 bps/yr	Notional × 0.05% × T
Brokerage	$2/lot	(Barrels / 1000) × $2

Total Cost = Option Cost + Brokerage + Slippage + Structuring + Distribution + CVA

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```python
class BlackScholes:
    @staticmethod
    def call_price(S, K, r, T, sigma):
        if T <= 1e-10: return max(S-K, 0)
        if sigma <= 1e-10: return max(S-K*np.exp(-r*T), 0)
        d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
        d2 = d1 - sigma*np.sqrt(T)
        return S*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d2)

    @staticmethod
    def strike_from_delta(S, delta, r, T, sigma):
        if T <= 1e-10 or sigma <= 1e-10: return S
        d1_target = norm.ppf(delta)
        ln_K = np.log(S) + (r+0.5*sigma**2)*T - d1_target*sigma*np.sqrt(T)
        return np.exp(ln_K)

    @staticmethod
    def delta(S, K, r, T, sigma):
        if T <= 1e-10: return 1.0 if S > K else 0.0
        d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
        return norm.cdf(d1)

    @staticmethod
    def gamma(S, K, r, T, sigma):
        if T <= 1e-10 or sigma <= 1e-10: return 0.0
        d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
        return norm.pdf(d1)/(S*sigma*np.sqrt(T))

    @staticmethod
    def vega(S, K, r, T, sigma):
        if T <= 1e-10 or sigma <= 1e-10: return 0.0
        d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
        return S*norm.pdf(d1)*np.sqrt(T)/100
```

```python
@staticmethod
def theta(S, K, r, T, sigma):
    if T <= 1e-10 or sigma <= 1e-10: return 0.0
    d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    t1 = -S*norm.pdf(d1)*sigma/(2*np.sqrt(T))
    t2 = -r*K*np.exp(-r*T)*norm.cdf(d2)
    return (t1+t2)/365

@staticmethod
def rho(S, K, r, T, sigma):
    if T <= 1e-10: return 0.0
    d2 = (np.log(S/K)+(r-0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    return K*T*np.exp(-r*T)*norm.cdf(d2)/100

@staticmethod
def vanna(S, K, r, T, sigma):
    if T <= 1e-10 or sigma <= 1e-10: return 0.0
    d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return -norm.pdf(d1)*d2/sigma

@staticmethod
def charm(S, K, r, T, sigma):
    if T <= 1e-10 or sigma <= 1e-10: return 0.0
    d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return norm.pdf(d1)*(2*r*T - d2*sigma*np.sqrt(T))/(2*T*sigma*np.sq

@staticmethod
def vomma(S, K, r, T, sigma):
    if T <= 1e-10 or sigma <= 1e-10: return 0.0
    d1 = (np.log(S/K)+(r+0.5*sigma**2)*T)/(sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    vega = S*norm.pdf(d1)*np.sqrt(T)/100
    return vega*d1*d2/sigma

        @staticmethod
        def all_greeks(S, K, r, T, sigma):
            return {
                'delta': BlackScholes.delta(S,K,r,T,sigma),
                'gamma': BlackScholes.gamma(S,K,r,T,sigma),
                'vega': BlackScholes.vega(S,K,r,T,sigma),
                'theta': BlackScholes.theta(S,K,r,T,sigma),
                'rho': BlackScholes.rho(S,K,r,T,sigma),
                'vanna': BlackScholes.vanna(S,K,r,T,sigma),
                'charm': BlackScholes.charm(S,K,r,T,sigma),
                'vomma': BlackScholes.vomma(S,K,r,T,sigma)
            }

bs = BlackScholes()
print('Black-Scholes engine with 8 Greeks ready')
```

Black-Scholes Model

Call Option Price:

$$C = S \cdot N(d_1) - K \cdot e^{-rT} \cdot N(d_2)$$

Where:

$$d_1 = \frac{\ln(S/K) + (r + \frac{\sigma^2}{2})T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

Strike Calculation from Delta

To find strike K for a target delta Δ:

$$K = S \cdot \exp\left[(r + \frac{\sigma^2}{2})T - N^{-1}(\Delta) \cdot \sigma\sqrt{T}\right]$$

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```python
def run_single_backtest(market_data, inception_date, product, costs, cost_calc
    try:
        row = market_data.loc[inception_date]
    except:
        return None

    S0, sig0, r0 = row['spot'], row['vol'], row['rate']
    if S0 <= 0 or sig0 <= 0.05:
        return None

    future = market_data.index[market_data.index > inception_date]
    if len(future) < product.tenor_days:
        return None

    expiry = future[min(product.tenor_days-1, len(future)-1)]
    daily = market_data.loc[inception_date:expiry].copy()
    if len(daily) < 50:
        return None

    T0 = len(daily) / 252

    K_long = bs.strike_from_delta(S0, product.long_call_delta, r0, T0, sig0)
    K_short = bs.strike_from_delta(S0, product.short_call_delta, r0, T0, sig0)
    if K_short <= K_long:
        K_long, K_short = S0*1.00, S0*1.20

    c_long = bs.call_price(S0, K_long, r0, T0, sig0)
    c_short = bs.call_price(S0, K_short, r0, T0, sig0)
    spread_cost = c_long - c_short
    if spread_cost <= 0.01:
        return None

    if product.use_fixed_participation:
        participation = product.target_participation
        max_spread = K_short - K_long
        normal_spread_pct = 0.1875
        capped_max_spread = min(max_spread, S0 * normal_spread_pct)
        barrels = (product.notional * participation) / S0
        # Store the capped max spread for later use
        max_spread_capped = capped_max_spread
```

```python
    else:
        zcb_cost = product.notional / ((1+costs.issuer_funding_rate)**T0)
        barrels = (product.notional - zcb_cost) / spread_cost
        participation = barrels*(K_short-K_long) / (product.notional*(K_short-K_long)/S0)
        max_spread_capped = K_short - K_long

    all_costs = cost_calc.total_costs(product.notional, spread_cost, barrels, T0)

    # Spot max drawdown
    spot_series = daily['spot']
    spot_cummax = spot_series.cummax()
    spot_drawdown = (spot_series - spot_cummax) / spot_cummax * 100
    spot_max_dd = spot_drawdown.min()

    marks = []
    for idx, (date, row) in enumerate(daily.iterrows()):
        S, sig, r = row['spot'], row['vol'], row['rate']
        if S <= 0 or sig <= 0.05:
            continue

        days_left = len(daily) - idx - 1
        T = max(days_left/252, 1e-10)
        is_exp = (days_left == 0)

        if is_exp:
            cl, cs = max(S-K_long, 0), max(S-K_short, 0)
            zcb = product.notional
            greeks = {k: 0 for k in ['delta','gamma','vega','theta','rho','vanna','charm','vomma']}
        else:
            cl = bs.call_price(S, K_long, r, T, sig)
            cs = bs.call_price(S, K_short, r, T, sig)
            zcb = product.notional / ((1+costs.issuer_funding_rate)**T)
            gl = bs.all_greeks(S, K_long, r, T, sig)
            gs = bs.all_greeks(S, K_short, r, T, sig)
            greeks = {k: (gl[k]-gs[k])*barrels for k in gl.keys()}

        spread = cl - cs
        # Cap spread at the maximum allowed spread width
        spread = min(spread, max_spread_capped)
        opt_val = barrels * spread
        # Also cap opt_val to ensure max return is ~13%
        max_opt_val = product.notional * participation * 0.1875
        opt_val = min(opt_val, max_opt_val)
        gross = zcb + opt_val
        pct_elapsed = idx / len(daily)
        costs_td = all_costs['total'] * (0.5 + 0.5*pct_elapsed)
        net = gross - costs_td
        pnl = net - product.notional + product.notional * 0.005

        marks.append({
            'date': date, 'days_left': days_left, 'spot': S, 'vol': sig,
            'spread': spread, 'opt_val': opt_val, 'zcb': zcb,
            'gross': gross, 'costs': costs_td, 'net': net,
            'pnl': pnl, 'pnl_pct': pnl/product.notional*100, **greeks
        })
```

Position Sizing

Number of Barrels:

$$\text{Barrels} = \frac{\text{Notional} \times \text{Participation}}{S_0}$$

Maximum Return (Capped):

$$\text{Max Return} = \text{Participation} \times \min\left(\frac{K_2 - K_1}{S_0}, 20\%\right)$$

Daily Mark-to-Market

For each day t during the holding period:

$$\text{ZCB}_t = \frac{\text{Notional}}{(1 + r_{\text{funding}})^{T_{\text{remaining}}}}$$

$$\text{Spread}_t = C(S_t, K_1) - C(S_t, K_2)$$

$$\text{Option Value}_t = \text{Barrels} \times \min(\text{Spread}_t, \text{Max Spread})$$

$$\text{Gross}_t = \text{ZCB}_t + \text{Option Value}_t$$

$$\text{Net}_t = \text{Gross}_t - \text{Costs}_t$$

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```python
def run_rolling_backtest(market_data, product, costs, cost_calc, bt_cfg):
    print('')
    print('='*60)
    print('RUNNING ROLLING BACKTEST')
    print('='*60)

    end_idx = len(market_data) - product.tenor_days - 10
    if end_idx <= 0:
        print('ERROR: Not enough data')
        return None, None

    dates = market_data.index[0:end_idx:bt_cfg.sample_frequency]
    print('Testing {} inception dates'.format(len(dates)))
    print('From {} to {}'.format(dates[0].strftime('%Y-%m-%d'), dates[-1].strftime('%Y-%m-%d')))

    results, full = [], []
    for i, d in enumerate(dates):
        if i % 100 == 0:
            print('Progress: {}/{}'.format(i, len(dates)))

        r = run_single_backtest(market_data, d, product, costs, cost_calc)
        if r:
            full.append(r)
            results.append({
                'inception': r['inception'], 'expiry': r['expiry'],
                'S0': r['S0'], 'final_spot': r['final_spot'],
                'spot_ret': r['spot_ret'], 'spot_max_dd': r['spot_max_dd'],
                'K_long': r['K_long'], 'K_short': r['K_short'],
                'participation': r['participation'], 'costs_pct': r['costs_pct'],
                'gross_ret': r['gross_ret'], 'net_ret': r['net_ret'],
                'is_zero': r['is_zero'], 'is_capped': r['is_capped'],
                'max_dd': r['risk']['max_dd'], 'sharpe': r['risk']['sharpe'],
                'sortino': r['risk']['sortino'],
                'var_95': r['risk']['var_95'], 'cvar_95': r['risk']['cvar_95'],
                'volatility': r['risk']['volatility']
            })

    print('')
    print('Completed {} backtests'.format(len(results)))
    return pd.DataFrame(results), full

summary_df, full_results = run_rolling_backtest(market_data, product, costs, cost_calc, bt_cfg)
```

Rolling Backtest Methodology

Process:

1. Sample inception dates every 5 business days from 2015-2025
2. For each inception date:
    - Calculate strikes based on that day's spot, vol, rates
    - Run 12-month holding period simulation
    - Track daily MTM and final payoff
    - Calculate risk metrics
3. Aggregate results across all 526 backtests

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```python
rets = df['pnl_pct'].diff().dropna()
neg_rets = rets[rets < 0]

risk = {
    'var_95': np.percentile(rets, 5) if len(rets) > 0 else 0,
    'cvar_95': neg_rets.mean() if len(neg_rets) > 0 else 0,
    'max_dd': df['pnl_pct'].min(),
    'max_gain': df['pnl_pct'].max(),
    'volatility': rets.std() * np.sqrt(252) if len(rets) > 0 else 0,
    'sharpe': (rets.mean()/rets.std()*np.sqrt(252)) if rets.std() > 0 else 0,
    'sortino': (rets.mean()/neg_rets.std()*np.sqrt(252)) if len(neg_rets) > 0 and neg_rets.std() > 0 else 0,
}
```

```python
def run_monte_carlo(S0, K_long, K_short, r, T, sigma, num_paths, participation):
    np.random.seed(None)
    Z = np.random.standard_normal(num_paths)
    ST = S0*np.exp((r-0.5*sigma**2)*T + sigma*np.sqrt(T)*Z)

    long_payoff = np.maximum(ST-K_long, 0)
    short_payoff = np.maximum(ST-K_short, 0)
    spread = long_payoff - short_payoff
    returns = (spread/S0)*participation*100 + 0.5

    is_zero = (ST <= K_long)
    is_capped = (ST >= K_short)

    df = pd.DataFrame({'final_spot':ST,'spread_payoff':spread,'return_pct':returns,
                       'is_zero':is_zero,'is_capped':is_capped})

    neg_rets = returns[returns < 0]
    summary = {
        'mean': returns.mean(),
        'median': np.median(returns),
        'std': returns.std(),
        'var_95': np.percentile(returns, 5),
        'cvar_95': neg_rets.mean() if len(neg_rets) > 0 else 0,
        'prob_zero': is_zero.mean()*100,
        'prob_capped': is_capped.mean()*100,
        'prob_positive': (returns > 0).mean()*100
    }
    return df, summary

print('Monte Carlo ready')
```

Rolling Backtest Methodology

Metric	Formula
Sharpe Ratio	$\frac{\mu_{\text{daily}}}{\sigma_{\text{daily}}} \times \sqrt{252}$
Sortino Ratio	$\frac{\mu_{\text{daily}}}{\sigma_{\text{downside}}} \times \sqrt{252}$
Max Drawdown	$min_t\left(\frac{V_t - V_{\text{peak}}}{V_{\text{peak}}}\right)$
95% VaR	5th percentile of daily returns
95% CVaR	Mean of returns below VaR

Monte Carlo Simulation

GBM Price Path:

$$S_T = S_0 \cdot \exp\left[(r - \frac{\sigma^2}{2})T + \sigma\sqrt{T} \cdot Z\right]$$

Where $Z \sim N(0,1)$

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

Monte Carlo Simulation

Parameter	Value	Description
Number of Paths	10,000	Simulated price trajectories
Time Horizon (T)	1 year	252 trading days
Initial Spot (S_0)	Market data	Latest Brent price from dataset
Volatility (σ)	Market data	Latest 1Y ATM implied vol
Risk-Free Rate (r)	Market data	Latest 1Y Treasury yield
Long Strike (K_1)	Avg from backtest	50-Delta (ATM)
Short Strike (K_2)	Avg from backtest	25-Delta (OTM)
Participation (p)	63%	Fixed participation rate

Outputs Calculated:

Metric	Formula
Mean Return	$\mathbb{E}[\text{Return}]$
Median Return	50th percentile
Std Dev	$\sqrt{\text{Var}[\text{Return}]}$
95% VaR	5th percentile
95% CVaR	$\mathbb{E}[\text{Return} \mid \text{Return} < \text{VaR}]$
Prob(Zero)	$P(S_T \leq K_1)$
Prob(Capped)	$P(S_T \geq K_2)$
Prob(Positive)	$P(\text{Return} > 0)$

Price Evolution (Geometric Brownian Motion):

$$S_T = S_0 \cdot \exp\left[(r - \frac{\sigma^2}{2})T + \sigma\sqrt{T} \cdot Z\right]$$

Where $Z \sim N(0,1)$

Appendix 1.2: Quantitative Framework for Pricing and Simulation (cont.)

```
def run_stress_tests(S0, K_long, K_short, r, T, sigma, barrels, notional):
    scenarios = {
        'Base Case': (0, 0, 0, 'Current conditions'),
        '2014-15 Oil Crash': (-0.50, 0.50, -0.01, 'Oil collapse'),
        'COVID Crash (Mar 2020)': (-0.65, 1.00, -0.02, 'Pandemic shock'),
        'Ukraine War Spike': (0.40, 0.30, 0.01, 'Geopolitical'),
        'Moderate Decline -20%': (-0.20, 0.15, 0, 'Demand slowdown'),
        'Moderate Rally +20%': (0.20, 0, 0, 'Supply disruption'),
        'Strong Rally +35%': (0.35, 0.10, 0.01, 'Major disruption'),
        'Vol Spike Only': (0, 0.50, 0, 'Uncertainty'),
        'Vol Crush': (0, -0.30, 0, 'Market calms'),
        'Rate Hike +200bp': (-0.05, 0.05, 0.02, 'Fed tightening'),
        'Rate Cut -100bp': (0.05, 0.10, -0.01, 'Easing'),
        'Perfect Storm': (-0.40, 0.80, 0.02, 'Worst case')
    }

    results = []
    for name, (spot_sh, vol_sh, rate_sh, desc) in scenarios.items():
        S_s = S0*(1+spot_sh)
        sig_s = np.clip(sigma*(1+vol_sh), 0.10, 1.50)
        r_s = max(r+rate_sh, 0.001)

        cl = bs.call_price(S_s, K_long, r_s, T, sig_s)
        cs = bs.call_price(S_s, K_short, r_s, T, sig_s)
        spread = cl - cs
        opt_val = barrels * spread
        zcb = notional / ((1+r_s)**T)
        total = zcb + opt_val
        pnl = total - notional

        results.append({
            'scenario': name, 'description': desc,
            'spot_shock': spot_sh*100, 'vol_shock': vol_sh*100,
            'rate_shock': rate_sh*10000, 'pnl_pct': pnl/notional*100
        })

    return pd.DataFrame(results)
```

Stress Testing

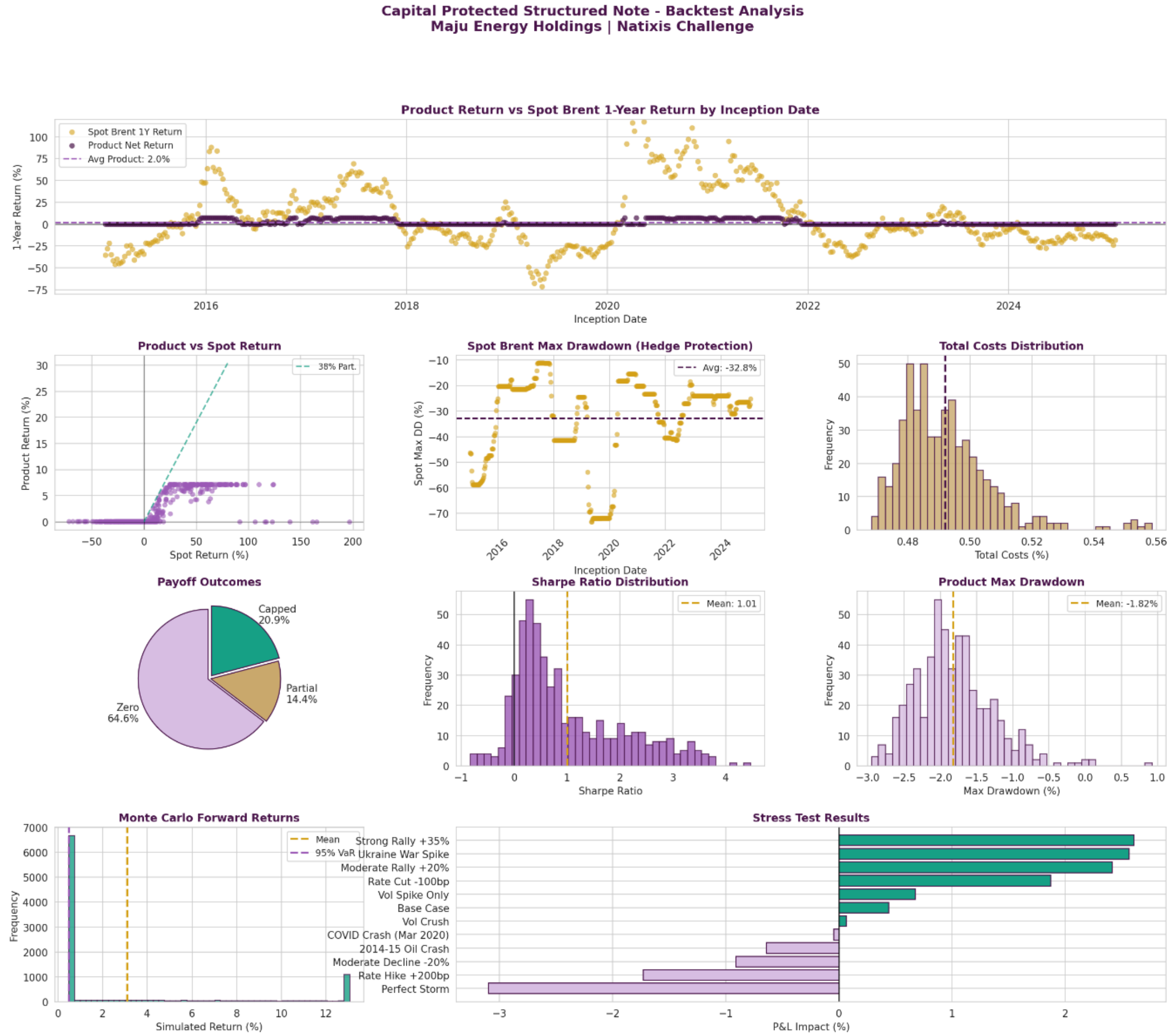
- Spot shock applied to initial spot price: $S_{stressed} = S_0 \times (1 + \text{shock})$
- Vol shock applied to initial implied vol: $\sigma_{stressed} = \sigma_0 \times (1 + \text{shock})$, clipped to [10%, 150%]
- Rate shock added to risk-free rate: $r_{stressed} = r_0 + \text{shock}$, floored at 0.1%

Stress P&L Calculation:

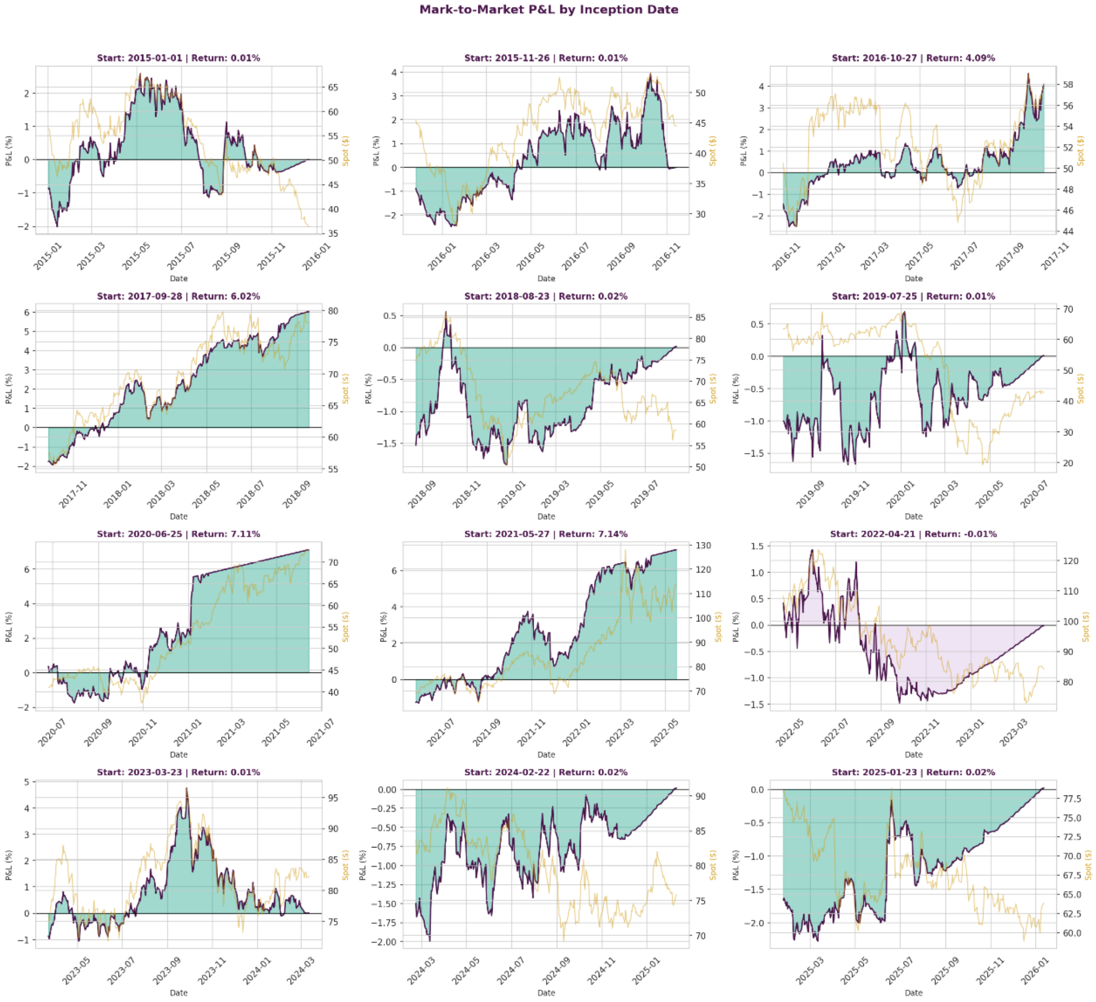
$$P\&L_{stress} = \frac{ZCB_{stressed} + Option_{stressed} - Notional}{Notional}$$

Appendix 1.3: Quantitative Framework for Backtesting Analysis

Chart Summary



MTM Performance of the Note for Difference Inception Dates



Appendix 2.1: Indicative Term Sheet

INDICATIVE TERM SHEET

Principal Protected Structured Note Linked to Brent Crude Oil

PRODUCT OVERVIEW

Issuer	Natixis Structured Issuance S.A.
Issuer Rating	S&P;: A+ / Fitch: A+
Investor	Maju Energy Holdings Pte Ltd
Notional Amount	USD 100,000,000
Currency	USD
Trade Date	15 January 2026
Issue Date (Asian Delay)	29 January 2026
Maturity Date	28 January 2027 (12 months)
Principal Protection	100% at Maturity
Fixed Coupon	0.50% p.a. (paid at maturity, subject to hold-to-maturity)

UNDERLYING REFERENCE

Underlying	ICE Brent Crude Oil Futures (Mar-27 – COH27)
Initial Fixing	15 January 2026 at Open
Final Fixing	28 January 2027 at Close
Initial Spot Price (S_0)	USD 65.26 (indicative)

OPTION STRUCTURE

Structure Type	Bull Call Spread
Option Style	European
Long Call Strike (K_1)	USD 65.00 (99.6% – 50 Delta, ATM)
Short Call Strike (K_2)	USD 77.50 (118.75% – 25 Delta, OTM)
Participation Rate	38%

COSTS & FEES

Issuer Funding Rate	3.89% p.a. (Natixis 1Y USD)
Structuring Fee	25 bps (one-time)
Distribution Fee	15 bps (one-time)
Total Estimated Costs	~0.52% of Notional

KEY RISK FACTORS

Credit Risk	Principal protection subject to Issuer creditworthiness
Market Risk	MTM volatility during holding period; no guaranteed return above principal
Liquidity Risk	Limited secondary market; early termination may result in loss
Currency Risk	Note is denominated in USD; investors with non-USD base currency are exposed to FX movements which may erode returns or principal when converted
Put Exercise Risk	If Investor exercises the Put, only 100% principal is returned – Fixed Coupon and any participation upside are forfeited
Opportunity Cost	If Brent ≤ K■ at maturity, investor receives principal plus Fixed Coupon only (no participation return)

IMPORTANT NOTES

- This is an indicative term sheet for discussion purposes only.
- Final terms subject to market conditions at time of execution.
- Options on futures reference the front-month contract at expiration.
- Product details shown as spot; industry norm for client understanding.
- Fixed Coupon of 0.50% is only payable if the Note is held to maturity and the Put is not exercised.

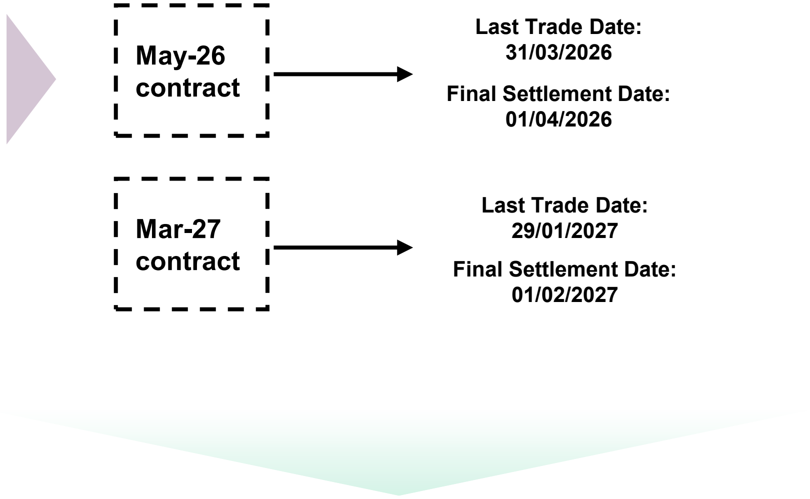
NATIXIS CORPORATE & INVESTMENT BANKING

Natixis Investment Strategy Challenge 2026

Appendix 2.2: Product Lifecycle (Assuming note held to maturity)

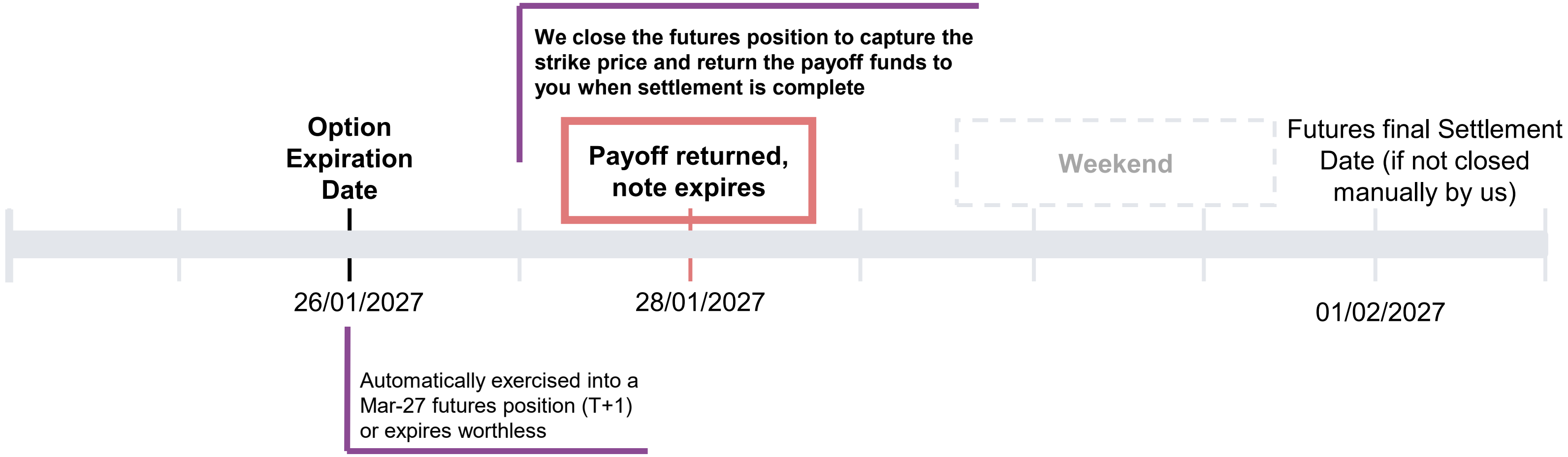
Understanding Commodity Futures and Option Expiration Convention

- Trading of a futures contract stops on the **last business day** of the **second month** prior to the delivery month.
- On the final settlement date, the **official cash settlement price** is calculated and applied - next business day after the last trade date.
- Options on futures expire 3 business days before the last trading day of the futures.



Option Expiry Case 1 – Futures expire on a Monday, Tuesday or Wednesday

Expiration Conventions for your Product



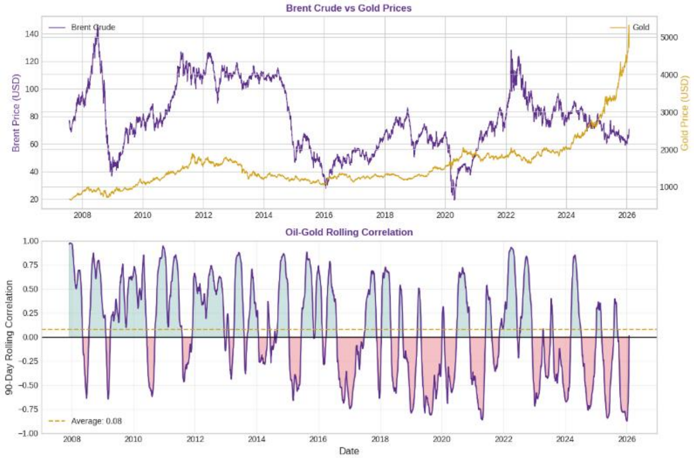
Case 1: Both options expire ITM
- The delivery right/obligation is netted out by the exchange, we are net flat futures and lock in the cash P&L to be distributed to you.

Case 2: The long call expires ITM, the short call OTM
- The long call gives us the right to exercise into the futures contract at the strike price

Case 3: Both options expire OTM
- No cash or futures transactions occur and the options contracts simple seize existing

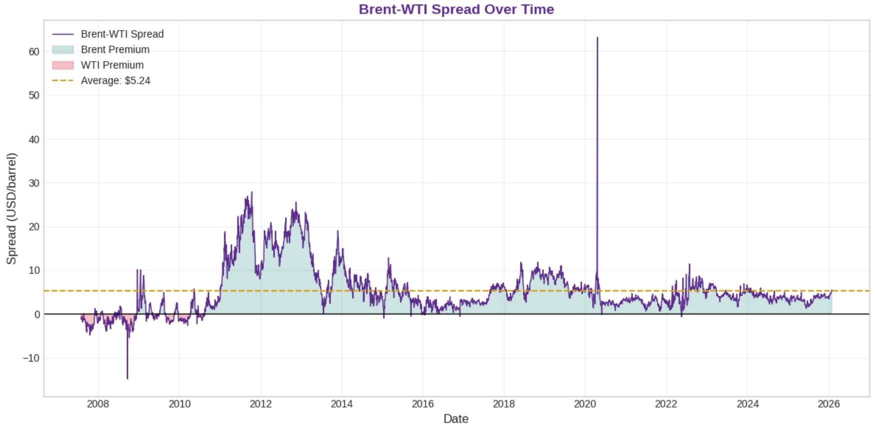
Appendix 2.3: Brent-related Correlation Analysis

Oil-Gold correlation over time



- Long-term correlation: 0.075
- Current 90D correlation: 0.011

Brent-WTI spread analysis

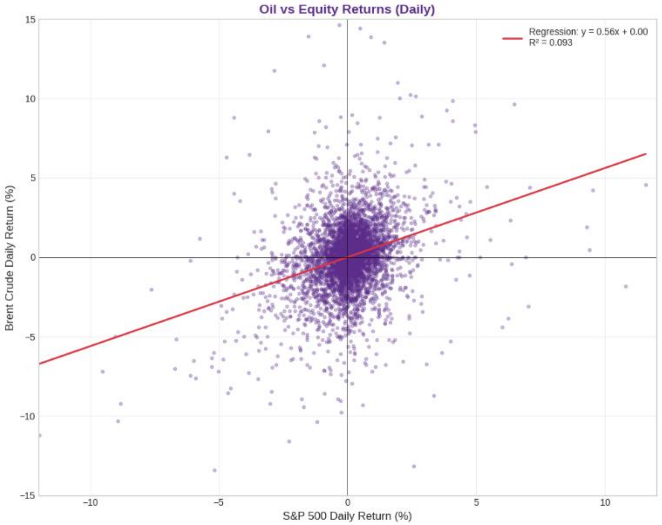


- Historical avg spread: $5.24
- Current spread: $5.48
- Max spread: $63.20

Multi-asset correlation heatmap

- Brent-WTI: 0.974
- Brent-Gold: -0.023
- Brent-S&P500: -0.194
- Brent-VIX: -0.131

Oil vs S&P 500 scatter plot with regression

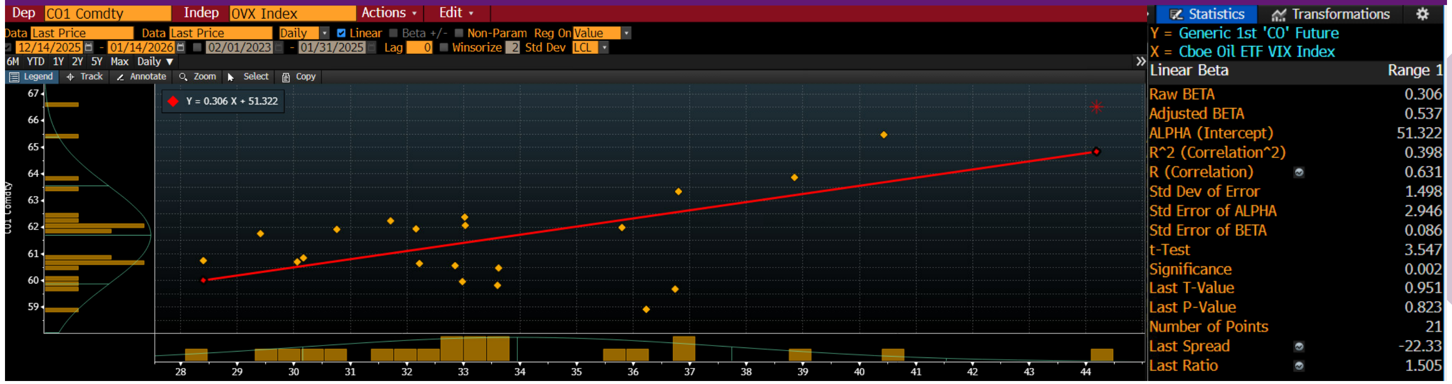


- Beta (slope): 0.561
- R-squared: 0.093
- Correlation: 0.304

Appendix 2.3: Brent-related Correlation Analysis (cont.) - Brent Futures and OVX

Source: Bloomberg <HRA>; Data Source: ICE Futures, CBOE

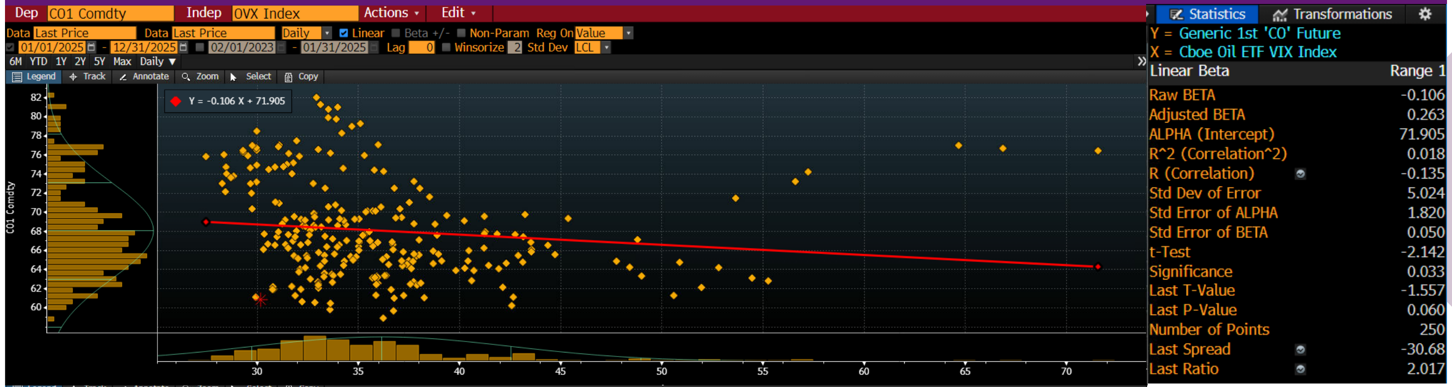
1M Lookback Correlation



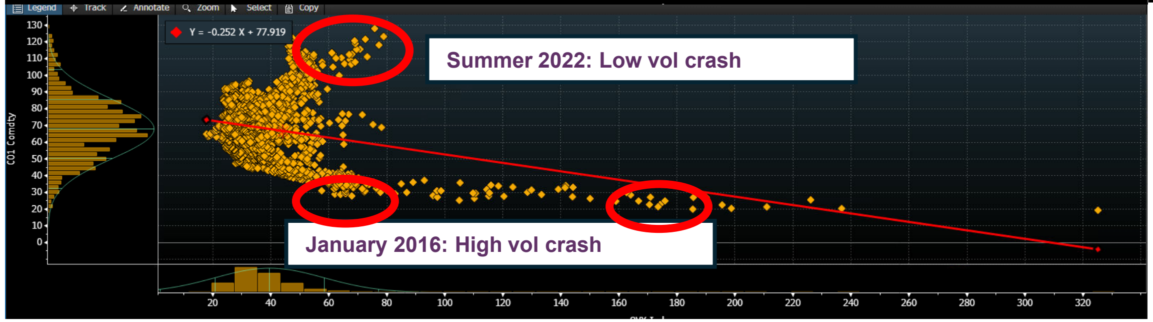
- Correlation Coefficient: **0.631**
- Low statistical significance due to **limited number of observations**
- Regardless, prices have **both moved higher** in January in response to the same headlines and geopolitical developments

-> due to the immense macroeconomic implications of supply-side oil shocks, **volatility occurs on moves to the upside** as opposed to during market crashes

2025 Correlation between active ICE Brent Futures Contract and CBOE OVX Index



- Correlation Coefficient: **-0.135**
- Significant t-test confirms the **slightly negative but insignificant correlation** between prices as volatility
- It shows that volatility, linked to panic buying and selling and sudden need for hedges can occur to the upside and downside in oil prices
- Still, as opposed to YTD volatility is more common to occur with crashes in price
- The same trend persists for longer lookback windows



Correlation from 14/01/2016 – 14/01/2026

- Correlation Coefficient: **-0.267**
- **Historic correlation between Brent oil and volatility is slightly negative**
- Outlier periods circled in the chart:

1. **Summer 2022:** After a strong rally linked to Russia's invasion of Ukraine prices pulled back on relatively low volatility
2. **Jan 2016 & April 2020:** Prices crashed linked to huge oversupply in the market mixed with very high volatility

Appendix 2.4: Call Spread Analysis

Visualising call skew for different expiration dates

IV Difference - July (6M)

	ATM	35DC	25DC	15DC	10DC
ATM	0	-0.26	0.01	1.16	2.89
35DC	0.26	0	0.27	1.42	3.15
25DC	-0.01	-0.27	0	1.15	2.88
15DC	-1.16	-1.42	-1.15	0	1.73
10DC	-2.89	-3.15	-2.88	-1.73	0

IV Difference - Oct (9M)

	ATM	35DC	25DC	15DC	10DC
ATM	0	-0.67	-0.71	-0.02	1.57
35DC	0.67	0	-0.04	0.65	2.24
25DC	0.71	0.04	0	0.69	2.28
15DC	0.02	-0.65	-0.69	0	1.59
10DC	-1.57	-2.24	-2.28	-1.59	0

IV Difference - Jan 2027 (12M)

	ATM	35DC	25DC	15DC	10DC
ATM	0	-0.71	-0.78	-0.09	1.53
35DC	0.71	0	-0.07	0.62	2.24
25DC	0.78	0.07	0	0.69	2.31
15DC	0.09	-0.62	-0.69	0	1.62
10DC	-1.53	-2.24	-2.31	-1.62	0

Cost of call spreads by component

Call Option Costs by Delta

Jul-26

	Long ATM	Short Call	Net Cost
ATM-35DC	4.255	2.425	**1.83**
ATM-25DC	4.255	1.55	**2.705**
ATM-15DC	4.255	0.835	**3.42**
ATM-10DC	4.255	0.57	**3.685**

Oct-26

	Long ATM	Short Call	Net Cost
ATM-35DC	**4.97**	2.74	**2.23**
ATM-25DC	**4.97**	1.75	**3.22**
ATM-15DC	**4.97**	0.995	**3.975**
ATM-10DC	**4.97**	0.645	**4.325**

Jan-27

	Long ATM	Short Call	Net Cost
ATM-35DC	**6.18**	3.62	**2.56**
ATM-25DC	**6.18**	2.36	**3.82**
ATM-15DC	**6.18**	1.28	**4.9**
ATM-10DC	**6.18**	0.79	**5.39**

Participation rates without Putability

Participation Rate Adjustments

**0.5% min. Cpn assumed p.a.*

	6M
ATM-35DC	0.591224
ATM-25DC	0.399978
ATM-15DC	0.316357
ATM-10DC	0.293607

	9M
ATM-35DC	0.719848
ATM-25DC	0.498529
ATM-15DC	0.403840
ATM-10DC	0.371159

	12M
ATM-35DC	0.827055
ATM-25DC	0.554256
ATM-15DC	0.432094
ATM-10DC	0.392813

PR with Putability

	12M
ATM-35DC	0.572133
ATM-25DC	0.383419
ATM-15DC	0.298910
ATM-10DC	0.271736

Appendix 2.5: Issuance Wording – Principal Protection and Issuance Delay

In Singapore, the wording "capital protected" is strongly restricted

Regulatory perspective

- "Capital protected" implies **full, unconditional protection**
- This conflicts with structured products being unsecured obligations
- Even with 100% principal structures, **issuer credit risk** exists

„Principal Protected" instead

- Signals that protection applies only to the principal amount invested
- Protection is **conditional** on holding to maturity
- **Acknowledges issuer credit risk and legal disclaimers**

Regulatory intent

- Prevent investor **misinterpretation**
- Prevent false sense of security and **misjudgment** of credit risk
- Clear and **transparent** product disclosure

Legal Basis in Singapore

- Structured products are regulated as **capital markets products** under the **Securities and Futures Act (Cap. 289).**
- MAS conducts supervisory oversight of how structured products are marketed, sold and disclosed.
- Marketing must comply with term sheet, product details and complex product disclosure requirements to avoid misleading investors.

"Asian Delay"

Regulatory perspective

In many Asian markets, including Singapore, structured products are subject to a post-trade issuance delay, commonly referred to as an "Asian delay", typically ranging from 5–10 business days after trade date

Purpose

- Final legal documentation is completed
- Issuance conditions are verified
- Regulatory, operational, and internal approvals are finalized
- Internal controls around booking, hedging, and issuance confirmation

Difference to other jurisdictions

- Same day or next day issuance is not uncommon
- In Asia, all conditions related to issuance must be satisfied before issuance can occur
- This allows issuers in Asia to cancel or adjust issuance if market disruption or other events occur during the delay period

Regulatory intent

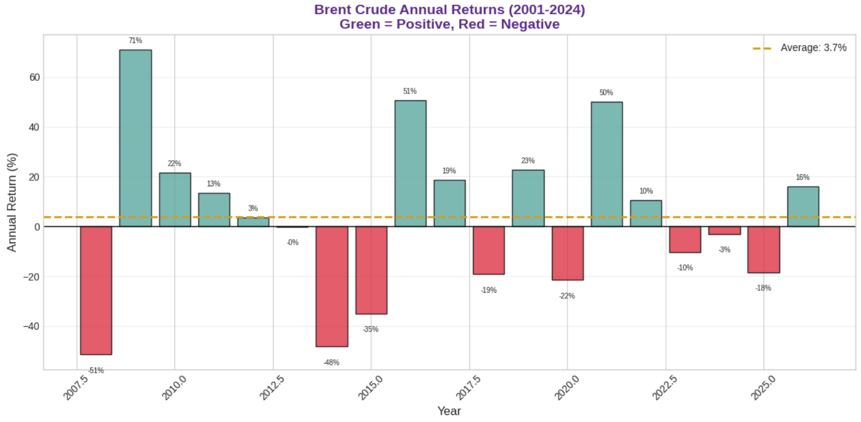
- Reduce operational and legal risk
- Ensure investors receive accurate final terms
- Prevent issuance of products under materially changed market conditions

Appendix 3.1: Historical Oil Price Analysis

Long-term Brent price history with key events (2000-2025)

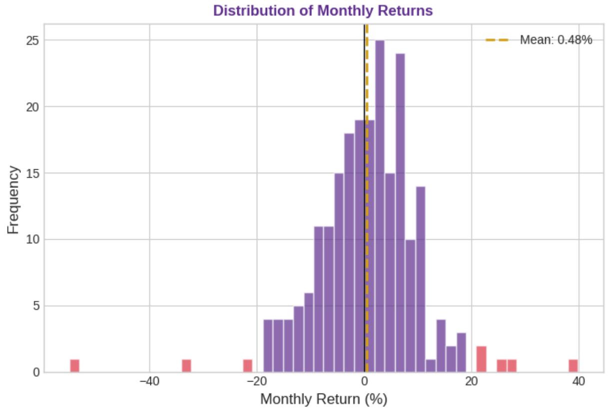
- Oil prices show 4-7 year super-cycles
- Supply shocks cause spikes (+40% to +120%)
- Demand shocks cause crashes (-60% to -80%)

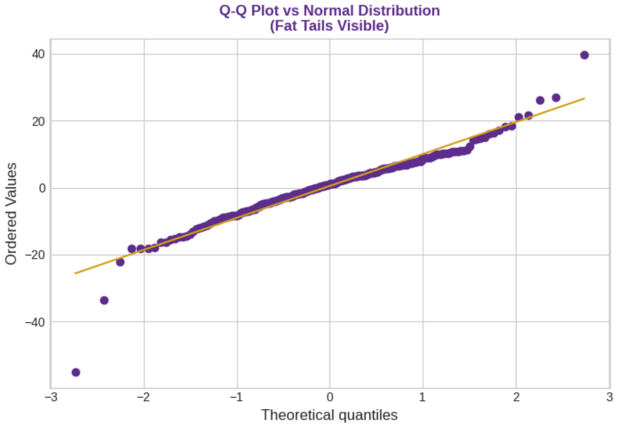
Annual returns distribution bar chart



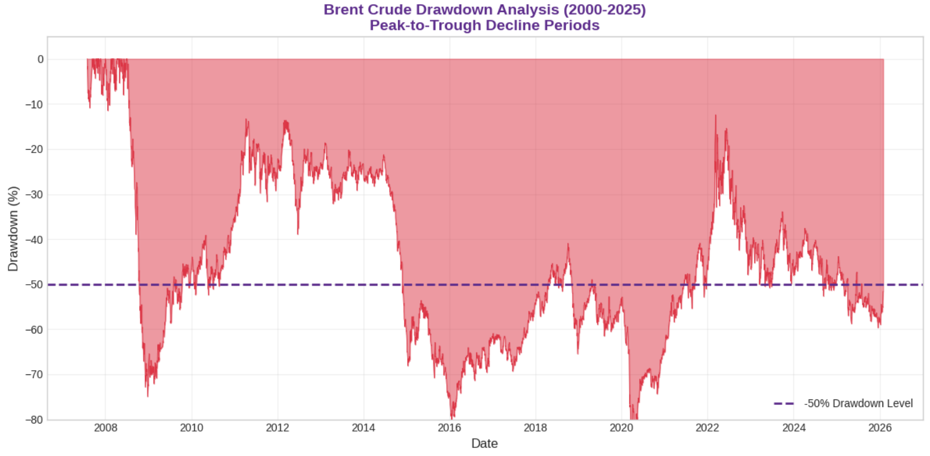
- Average annual return: 3.7%
- Std deviation: 32.7%
- Positive years: 10 (53%)
- Negative years: 9 (47%)

Monthly return distribution (histogram + Q-Q plot)





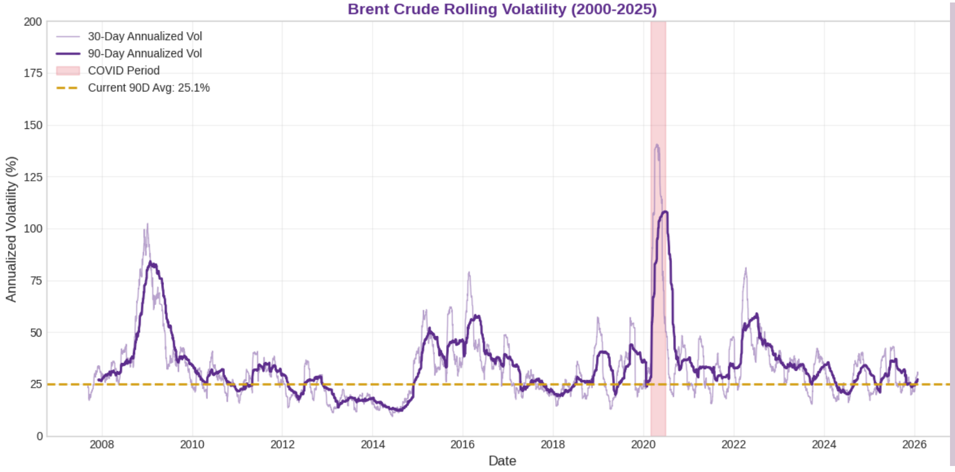
Drawdown analysis



- Maximum drawdown: -86.8%
- Days in >30% drawdown: 3412
- Days in >50% drawdown: 2063

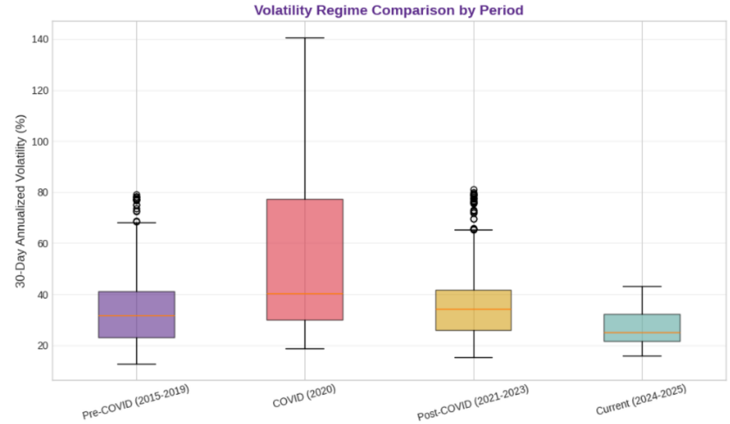
Appendix 3.2: Volatility Analysis (Pre/Post COVID)

Rolling volatility over time



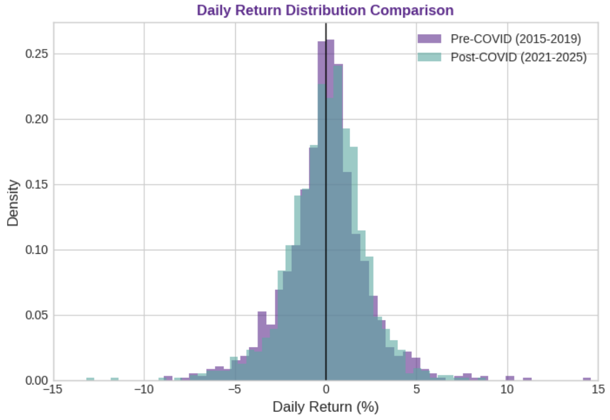
- Pre-COVID avg vol: 31.2%
- COVID period vol: 56.7%
- Post-COVID avg vol: 31.9%
- Current 30D vol: 29.3%

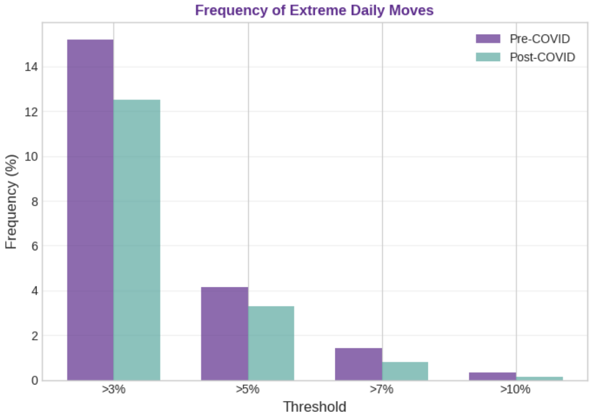
Pre vs Post COVID volatility box plot



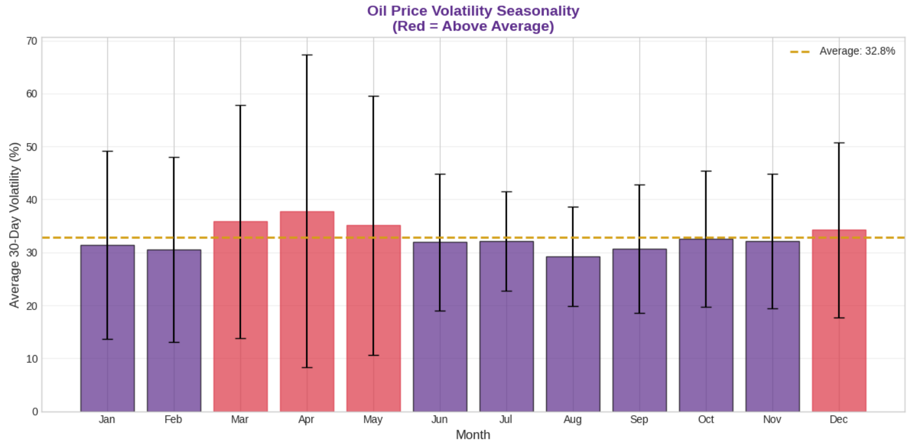
- Pre-COVID: Mean=33.7%, Median=31.7%
- COVID: Mean=56.7%, Median=40.4%
- Post-COVID: Mean=35.3%, Median=34.3%
- Current: Mean=27.1%, Median=25.1%

Daily return magnitude comparison





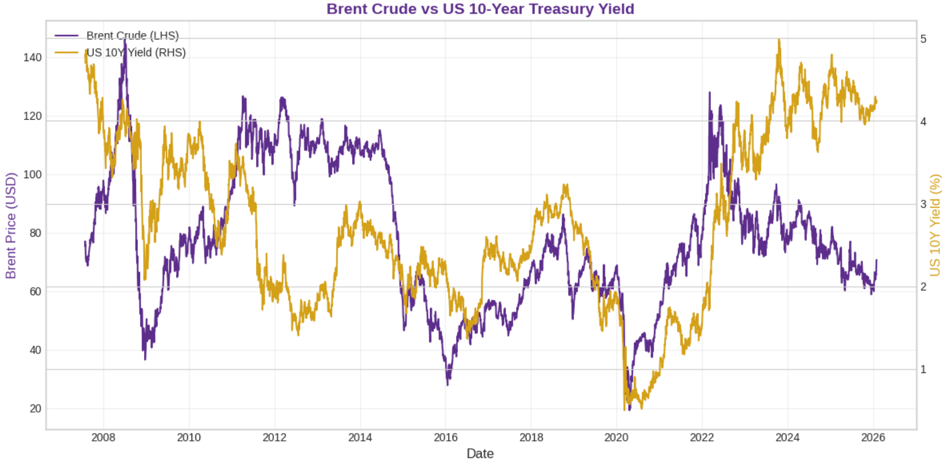
Volatility seasonality by month



- Highest vol month: Mar (37.8%)
- Lowest vol month: Jul (29.3%)

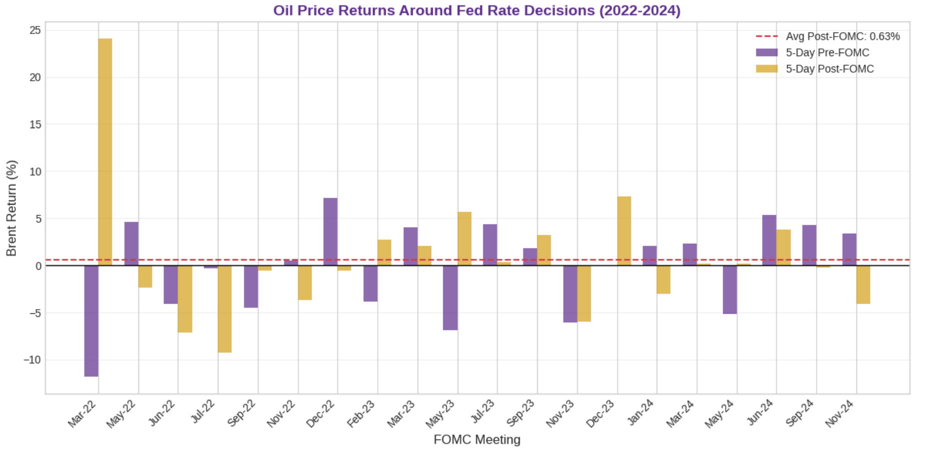
Appendix 3.3: Rate Decisions & Market Structure

Oil vs interest rates



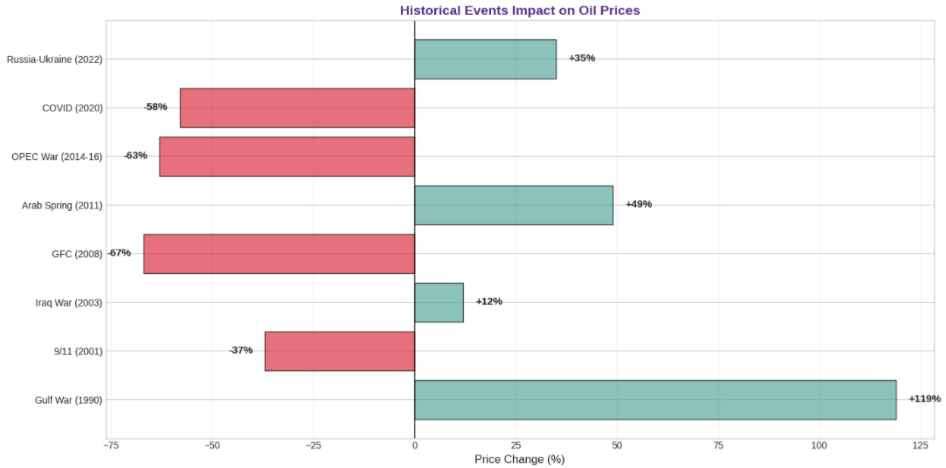
- Oil-Rate correlation: 0.275

Fed rate decision impact analysis



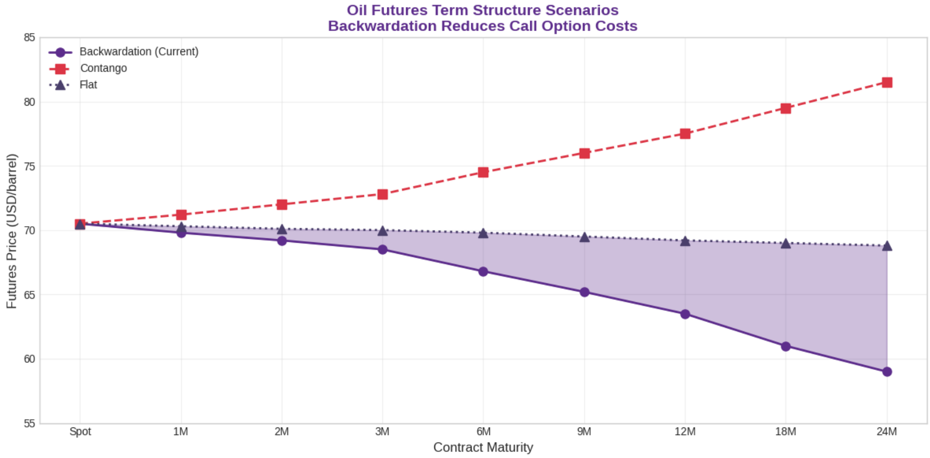
- Avg pre-FOMC return: -0.15%
- Avg post-FOMC return: 0.63%

Historical events impact on oil



- Supply shocks → price spikes (+35% to +119%)
- Demand shocks → price crashes (-37% to -67%)

Futures term structure (backwardation)



- Backwardation is present >90% of time
- 12M discount to spot: ~$7.0
- Lower forward = cheaper ATM calls